

# James Richard Forbes

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## Research Interests

### *State Estimation and Navigation*

Nonlinear state estimation, Kalman-like filters (e.g., extended Kalman filter (EKF), sigma point Kalman filter, invariant EKF), batch methods, variational inference, nonlinear deterministic observers, simultaneous localization and mapping (SLAM), estimation of systems with matrix Lie group states (e.g.,  $SO(3)$ ,  $SE(3)$ , homography), constrained state estimation, Koopman-based navigation, machine-learning-based approaches to enhance navigation.

### *Control Systems*

Robust control, optimal control, controller synthesis via linear matrix inequalities (LMIs),  $\mu$  analysis and synthesis, gain-scheduled control, input-output stability analysis and control, QSR dissipativity, control of systems with matrix Lie group states, Koopman-based control, optimization as a control problem.

### *Guidance*

Extremum seeking guidance, guidance of systems with matrix Lie group states.

### *Machine Learning and System Identification*

Applications of Gaussian-process regression, Koopman-operator-based system identification, time- and frequency-domain system identification.

### *Robotics and Aerospace Engineering Applications*

Multicopter and fixed-wing unmanned aerial vehicles (UAVs), autonomous underwater vehicles (AUVs), autonomous ground vehicles (AGVs) such as rail vehicles, on- and off-road vehicles, and spherical (rolling) robots, flexible robotic manipulators, spacecraft, cable-actuated robots, wind-energy-harvesting kites, system identification of vehicles.

Social Media: [decar.ca](https://decar.ca), Instagram, X/Twitter, LinkedIn, YouTube.

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## University Education

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| November 10, 2011 | <b>Doctor of Philosophy</b> , Aerospace Science and Engineering<br>University of Toronto Institute for Aerospace Studies<br>Toronto, ON, Canada<br>Thesis Title: <i>Extensions of Input-Output Stability Theory and the Control of Aerospace Systems</i><br>Supervisor: Prof. Christopher Damaren<br>Thesis committee: Prof. Timothy Barfoot and Prof. Hugh Liu<br>Winner of the 2011 G.N. Patterson Award for the most outstanding PhD thesis. |
| June 16, 2008     | <b>Master of Applied Science</b> , Aerospace Science and Engineering<br>University of Toronto Institute for Aerospace Studies<br>Toronto, ON, Canada                                                                                                                                                                                                                                                                                            |

Thesis Title: *Design of Optimal Strictly Positive Real Controllers Using Numerical Optimization for the Control of Large Flexible Space Structures*  
Supervisor: Prof. Christopher Damaren

June 17, 2006

**Bachelor of Applied Science**, Honours Mechanical Engineering, Co-op  
University of Waterloo  
Waterloo, ON, Canada  
Dean's Honours, with distinction.  
Fourth-Year Project Title: *Design of a Linear Electromagnetic Actuator*  
Fourth-Year Project Supervisor: Prof. Behrad Khamesee

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## Professional Appointments, Affiliations, Qualifications, and Memberships

### Academic Appointments

May 1, 2025 - Present

Professor  
Dept. of Mechanical Engineering, McGill University  
Macdonald Engineering Building  
817 Sherbrooke St. West, Montreal, QC, Canada, H3A 0C3

June 1, 2018 - May 1, 2025

Associate Professor  
Dept. of Mechanical Engineering, McGill University  
Macdonald Engineering Building  
817 Sherbrooke St. West, Montreal, QC, Canada, H3A 0C3

April 30, 2018 - Present

William Dawson Scholar, McGill University

August 1, 2015 - May 30, 2018 &  
May 16, 2011 - August 31, 2013

Assistant Professor  
Dept. of Mechanical Engineering, McGill University  
Macdonald Engineering Building  
817 Sherbrooke St. West, Montreal, QC, Canada, H3A 0C3

September 1, 2013 - June 30, 2015

Assistant Professor  
Dept. of Aerospace Engineering, University of Michigan  
FXB Building  
1320 Beal Avenue, Ann Arbor, MI, USA, 48109 - 2140

### Professional Affiliations

September 22, 2025 - Present

**Associate Academic Member**  
Mila - Quebec Artificial Intelligence Institute  
6666 Saint-Urbain Street, Montreal, QC, H2S 3H1

December 2, 2022 - Present

**Member**  
Trottier Institute for Sustainability in Engineering and Design (TISED)  
Macdonald Engineering Building  
McGill University  
817 Sherbrooke Street West, Montreal, QC, H3A 0C3

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|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| September 20, 2022 - Present | <b>Member</b><br>Centre de Recherches Mathématiques (CRM)<br>Pavillon André-Aisenstadt<br>2920 Chemin de la tour, Montréal, QC, Canada, H3T 1J4                                                                  |
| December 9, 2016 - Present   | <b>Member</b><br>Groupe d'études et de recherche en analyse des décisions (GERAD)<br>HEC Montréal<br>3000 Côte-Sainte-Catherine, Montréal, QC, Canada, H3T 2A7                                                   |
| October 16, 2015 - Present   | <b>Member</b><br>Regroupement pour l'étude des environnements partagés intelligents répartis (REPARTI)<br>Faculté des sciences et de génie - Pavillon Pouliot<br>Université Laval<br>Québec, QC, Canada, G1K 7P4 |
| October 16, 2015 - Present   | <b>Full Member</b><br>Centre for Intelligent Machines (CIM)<br>McConnell Engineering Building<br>McGill University<br>3480 University Street, Montreal, QC, Canada, H3A 0C3                                      |

## Professional Qualifications

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|-------------------------|-------------------------------------------------------------------------------------------------------------------|
| July 21, 2017 - Present | <b>Licensed Engineer</b><br>Professional Engineers of Ontario (PEO). License number: 100228385.                   |
| June 16, 2017 - Present | <b>Certificate of Proficiency in Amateur Radio (Basic)</b><br>Certificate number: 20170001007. Call sign: VE2VEO. |

## Professional Memberships

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|----------------|--------------------------------------------------------------------|
| 2011 - Present | Member, Institute of Electrical and Electronics Engineers (IEEE)   |
| 2012 - 2018    | Member, American Institute for Aeronautics and Astronautics (AIAA) |
| 2017 - 2018    | Member, American Society for Engineering Education (ASEE)          |
| 2012 - 2018    | Member, Canadian Aeronautics and Space Institute (CASI)            |

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## Publications

### Refereed Journal Publications

- [J1] M. Shalaby, S. S. Ahmed, N. Dahdah, C. Cossette, J. Le Ny, and J. R. Forbes, "MILUV: A Multi-UAV Indoor Localization dataset with UWB and Vision," *International Journal of Robotics Research*, 2026. To appear.
- [J2] L. Lortie, S. Dahdah, and J. R. Forbes, "Forward-Backward Extended DMD with an Asymptotic Stability Constraint," *Journal of Nonlinear Science*, vol. 36, no. 29, p. 28 pages, 2026.

- [J3] A. Stirling, M. Lukashchuk, D. Bagaev, W. Kouw, and J. R. Forbes, “Gaussian Variational Inference with Non-Gaussian Factors for State Estimation: A UWB Localization Case Study,” *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 2762–2769, 2026. Jointly accepted to ICRA.
- [J4] T. Adams and J. R. Forbes, “Design of Input-Output Observers for a Population of Systems with Bounded Frequency-Domain Variation using DK-iteration,” *IEEE Control Systems Letters*, vol. 9, pp. 2645–2650, 2025. Jointly accepted to ACC.
- [J5] S. Dahdah and J. R. Forbes, “pykoop: Python Library for Koopman Operator Approximation,” *Journal of Open Source Software*, vol. 10, no. 114, p. 7947, 2025.
- [J6] V. Korotkine, M. Cohen, and J. R. Forbes, “Globally Optimal Data-Association-Free Landmark-Based Localization Using Semidefinite Relaxations,” *IEEE Robotics and Automation Letters*, vol. 10, no. 10, pp. 10250–10257, 2025. Jointly accepted to ICRA.
- [J7] S. Moalemi and J. R. Forbes, “Matrix Scheduling of QSR-Dissipative Systems,” *IEEE Transactions on Automatic Control*, vol. 70, no. 8, pp. 5286–5300, 2025.
- [J8] D. Rimorov, J. R. Forbes, O. Tremblay, and R. Gagnon, “Robust Impedance Emulation Using Optimal Filter Approach for Power-Hardware-In-the-Loop Application,” *IEEE Open Journal of the Industrial Electronics Society*, vol. 6, pp. 158–169, 2025.
- [J9] M. Saavedra, S. Parkison, R. Arora, J. R. Forbes, and L. Paull, “The Harmonic Exponential Filter for Nonparametric Estimation on Motion Groups,” *IEEE Robotics and Automation Letters*, vol. 10, no. 2, pp. 2096–2103, 2025.
- [J10] L. Lortie and J. R. Forbes, “Asymptotically Stable Data-Driven Koopman Operator Approximation with Inputs using Total Extended DMD,” *Machine Learning: Science and Technology*, vol. 6, no. 1, p. 015003 (16 pages), 2025.
- [J11] K. Jung, T. Hitchcox, and J. R. Forbes, “An Adaptive Graduated Nonconvexity Loss Function for Robust Nonlinear Least Squares Solutions,” *IEEE Transactions on Robotics*, vol. 40, pp. 4207–4221, 2024. Jointly accepted to ICRA.
- [J12] Z. C. Guo, F. Dumbgen, J. R. Forbes, and T. Barfoot, “Data-Driven Batch Localization and SLAM Using Koopman Linearization,” *IEEE Transactions on Robotics*, vol. 40, pp. 3964–3983, 2024. Jointly accepted to ICRA.
- [J13] V. Korotkine, M. Cohen, and J. R. Forbes, “A Hessian for Gaussian Mixture Likelihoods in Nonlinear Least Squares,” *IEEE Robotics and Automation Letters*, vol. 9, no. 9, pp. 7891–7898, 2024. Jointly accepted to ICRA.
- [J14] C. C. Cossette, M. A. Shalaby, D. Saussié, and J. R. Forbes, “Decentralized State Estimation: An Approach using Pseudomeasurements and Preintegration,” *International Journal of Robotics Research*, vol. 43, no. 10, pp. 1573–1593, 2024.
- [J15] S. Dahdah and J. R. Forbes, “Closed-loop Koopman Operator Approximation,” *Machine Learning: Science and Technology*, vol. 5, no. 2, p. 025038 (19 pages), 2024.
- [J16] J. Arsenault, M. Cunniff, E. Tulsky, and J. R. Forbes, “Spatial Roles in Hockey Special Teams,” *Journal of Quantitative Analysis in Sports*, vol. 20, no. 3, pp. 235–250, 2024.
- [J17] T. Hitchcox and J. R. Forbes, “Laser-to-Vehicle Extrinsic Calibration in Low-Observability Scenarios for Subsea Mapping,” *IEEE Robotics and Automation Letters*, vol. 9, no. 4, pp. 3522–3529, 2024. Jointly accepted to ICRA40.
- [J18] M. A. Shalaby, C. C. Cossette, J. Le Ny, and J. R. Forbes, “Reducing Two-Way Ranging Variance by Signal-Timing Optimization,” *IEEE Transactions Aerospace and Electronic Systems*, vol. 60, no. 3, pp. 3718–3724, 2024.
- [J19] M. A. Shalaby, C. C. Cossette, J. Le Ny, and J. R. Forbes, “Multi-Robot Relative Pose Estimation and IMU Preintegration Using Passive UWB Transceivers,” *IEEE Transactions on Robotics*, vol. 40, pp. 2410–2429, 2024.

- [J20] A. Bajwa, C. C. Cossette, M. A. Shalaby, and J. R. Forbes, “DIVE: Deep Inertial-Only Velocity Aided Estimation for Quadrotors,” *IEEE Robotics and Automation Letters*, vol. 9, no. 4, pp. 3728–3734, 2024. Jointly accepted to IROS.
- [J21] A. Del Castillo Bernal, P. Decoste, and J. R. Forbes, “Bayesian Filtering for Homography Estimation,” *IEEE Robotics and Automation Letters*, vol. 8, no. 12, pp. 8216–8223, 2023. Jointly accepted to ICRA.
- [J22] H. Chorney, J. R. Forbes, and M. Driscoll, “System Identification and Simulation of Soft Tissue Force Feedback in a Spine Surgical Simulator,” *Computers in Biology and Medicine*, vol. 164, p. 107267 (8 pages), 2023.
- [J23] A. Al-Baali, T. Hitchcox, and J. R. Forbes, “Combining DVL-INS and Laser-based Loop Closures in a Batch Estimation Framework for Underwater Positioning,” *IEEE Journal of Oceanic Engineering*, vol. 48, no. 4, pp. 1096–1111, 2023.
- [J24] D. Lisus, M. Cohen, and J. R. Forbes, “Know What You Don’t Know: Consistency in Sliding Window Filtering with Unobservable States Applied to Visual-Inertial SLAM,” *IEEE Robotics and Automation Letters*, vol. 8, no. 6, pp. 3382–3389, 2023.
- [J25] T. Hitchcox and J. R. Forbes, “Improving Self-Consistency in Underwater Mapping through Laser-Based Loop Closure,” *IEEE Transactions on Robotics*, vol. 39, no. 3, pp. 1873–1892, 2023. Jointly accepted to ICRA.
- [J26] F. Ahmed, L. A. Sobiesiak, and J. R. Forbes, “Cascaded Model Predictive Control of a Tandem-Rotor Helicopter,” *IEEE Control Systems Letters*, vol. 7, pp. 1345–1350, 2023.
- [J27] M. Doff-Sotta, M. Cannon, and J. R. Forbes, “Spacecraft Energy Management using Convex Optimisation,” *Energy Conversion and Management: X*, vol. 16, p. 100325 (14 pages), 2022.
- [J28] S. Dahdah and J. R. Forbes, “System Norm Regularization Methods for Koopman Operator Approximation,” *Proceedings of the Royal Society A*, vol. 478, no. 2265, p. (21 pages), 2022.
- [J29] A. Bianco, R. Zonis, A.-M. Lauzon, J. R. Forbes, and G. Ijpma, “System Identification and Two-Degree-of-Freedom Control of Nonlinear, Viscoelastic Tissues,” *IEEE Transactions on Biomedical Engineering*, vol. 69, no. 12, pp. 3803–3811, 2022.
- [J30] T. Hitchcox and J. R. Forbes, “Mind the Gap: Norm-Aware Adaptive Robust Loss for Multivariate Least-Squares Problems,” *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 7116–7123, 2022. Jointly accepted to IROS.
- [J31] F. Ahmed, L. A. Sobiesiak, and J. R. Forbes, “Model Predictive Control of a Tandem-Rotor Helicopter with a Non-Uniformly Spaced Prediction Horizon,” *IEEE Control Systems Letters*, vol. 6, pp. 2828–2833, 2022. Jointly accepted to CDC.
- [J32] K. Lee and J. R. Forbes, “Position and Attitude Tracking Control Using CCW and SNI System Theory With Applications to Multi-agent Systems,” *Automatica*, vol. 139, p. 110203 (9 pages), 2022.
- [J33] T. D. Barfoot, J. R. Forbes, and G. M. D’Eleuterio, “Vectorial Parameterizations of Pose,” *Robotica*, vol. 40, no. 7, pp. 2409–2427, 2022.
- [J34] Z. C. Guo, V. Korotkine, J. R. Forbes, and T. D. Barfoot, “Koopman Linearization for Data-Driven Batch State Estimation of Control-Affine Systems,” *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 866–873, 2022. Jointly accepted to ICRA.
- [J35] D. Mirza-Hekmati, W. Heath, J. Apsley, and J. R. Forbes, “Down-Speeding Diesel Engines with Two-Stage Turbochargers: Analysis and Control Considerations,” *International Journal of Engine Research*, vol. 23, no. 1, pp. 78–89, 2022.
- [J36] D. Lisus, C. C. Cossette, M. Shalaby, and J. R. Forbes, “Heading Estimation Using Ultra-wideband Received Signal Strength and Gaussian Processes,” *IEEE Robotics and Automation Letters*, vol. 6, no. 4, pp. 8387–8393, 2021. Jointly accepted to IROS.

- [J37] M. Shalaby, C. C. Cossette, J. Le Ny, and J. R. Forbes, “Cascaded Filtering Using the Sigma Point Transformation,” *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4758–4765, 2021. Jointly accepted to ICRA. Best student paper award finalist (1 of 4 finalists).
- [J38] M. Shalaby, C. C. Cossette, J. R. Forbes, and J. Le Ny, “Relative Position Estimation in Multi-Agent Systems Using Attitude-Coupled Range Measurements,” *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4955–4961, 2021. Jointly accepted to ICRA.
- [J39] C. C. Cossette, M. Shalaby, D. Saussié, J. R. Forbes, and J. Le Ny, “Relative Position Estimation Between Two UWB Devices with IMUs,” *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4313–4320, 2021. Jointly accepted to ICRA.
- [J40] X. Shan, J. Angeles, and J. R. Forbes, “Design and Configuration of Folded Platonic Strapdowns of Biaxial MEMS Accelerometers,” *ASME Journal of Mechanical Design*, vol. 143, no. 2, pp. 025001 (1–6), 2020.
- [J41] M. Cohen, K. Abdulrahim, and J. R. Forbes, “Finite-Horizon LQR Control of Quadrotors on  $SE_2(3)$ ,” *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 5748–5755, 2020. Jointly accepted to IROS.
- [J42] N. van der Laan, M. Cohen, J. Arsenault, and J. R. Forbes, “The Invariant Rauch-Tung-Striebel Smoother,” *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 5067–5074, 2020. Jointly accepted to IROS.
- [J43] T. D. Barfoot, J. R. Forbes, and D. Yoon, “Exactly Sparse Gaussian Variational Inference with Application to Derivative-Free Batch Nonlinear State Estimation,” *International Journal of Robotics Research*, vol. 39, no. 13, pp. 1473–1502, 2020.
- [J44] M. Cohen and J. R. Forbes, “Navigation and Control of Unconventional VTOL UAVs in Forward-Flight with Explicit Velocity Wind Estimation,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 1151–1158, 2020. Jointly accepted to ICRA.
- [J45] C. C. Cossette, A. Walsh, and J. R. Forbes, “The Complex-Step Derivative Approximation on Matrix Lie Groups,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 906–913, 2020. Jointly accepted to ICRA.
- [J46] C. C. Cossette, J. R. Forbes, and D. Saussié, “Lagrangian Derivation of Variable-Mass Equations of Motion using an Arbitrary Attitude Parameterization,” *Journal of the Astronautical Sciences*, vol. 67, pp. 1206–1219, 2020.
- [J47] Z. Yin, B. Belzile, J. Angeles, and J. R. Forbes, “Elastodynamics of a Parallel Schönflies-motion Generator,” *Transactions of the Canadian Society for Mechanical Engineering*, vol. 44, no. 4, 2020.
- [J48] M. Turner and J. R. Forbes, “Conic Sector Analysis Using Integral Quadratic Constraints,” *International Journal of Robust and Nonlinear Control*, vol. 30, no. 2, pp. 741–755, 2020.
- [J49] R. J. Caverly, L. J. Bridgeman, R. Pates, and J. R. Forbes, “MIMO Nyquist Interpretation of the Large Gain Theorem,” *International Journal of Control*, vol. 93, no. 10, pp. 2326–2335, 2020.
- [J50] X. Shan, J. Angeles, and J. R. Forbes, “A Novel Capacitive Sensing Structure for Simultaneous Detection of Biaxial Low-g Acceleration in a Commercial MEMS Process,” *Microsystem Technologies*, vol. 25, no. 12, pp. 4475–4481, 2019.
- [J51] L. J. Bridgeman and J. R. Forbes, “Iterative  $\mathcal{H}_2$ -Conic Controller Synthesis,” *International Journal of Robust and Nonlinear Control*, vol. 29, no. 11, pp. 3701–3714, 2019.
- [J52] J. R. Forbes, “Synthesis of Strictly Positive Real  $\mathcal{H}_2$  Controllers Using Dilated LMIs,” *International Journal of Control*, vol. 92, no. 11, pp. 2584–2590, 2019.
- [J53] H. A. Godbole, R. J. Caverly, and J. R. Forbes, “Dynamic Modelling and Adaptive Control of a Flexible Cable-Driven Parallel Robot,” *ASME Journal of Dynamic Systems, Measurement and Control*, vol. 141, no. 10, p. 101002 (13 pages), 2019.

- [J54] R. Aucoin, S. A. Chee, and J. R. Forbes, “Nonlinear Approaches to Linear- and Linear-Matrix-Inequality-Constrained State Estimation,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. 55, no. 6, pp. 3153–3167, 2019.
- [J55] D. E. Zlotnik and J. R. Forbes, “Higher-Order Nonlinear Complementary Filtering on Lie Groups,” *IEEE Transactions on Automatic Control*, vol. 64, no. 5, pp. 1772–1783, 2019.
- [J56] D. E. Zlotnik and J. R. Forbes, “Gradient-Based Observer for Simultaneous Localization and Mapping,” *IEEE Transactions on Automatic Control*, vol. 63, no. 12, pp. 4338–4344, 2018.
- [J57] R. J. Caverly and J. R. Forbes, “ $\mathcal{H}_\infty$ -Optimal Parallel Feedforward Control Using Minimum Gain,” *IEEE Control Systems Letters*, vol. 2, no. 4, pp. 677–682, 2018.
- [J58] A. Walsh and J. R. Forbes, “Constrained Attitude Control on  $SO(3)$  via Semidefinite Programming,” *AIAA Journal of Guidance, Control, and Dynamics*, vol. 41, no. 11, pp. 2480–2485, 2018.
- [J59] X. Shan, J. Angeles, and J. R. Forbes, “Design of a Biaxial High Frequency Ratio Low-g MEMS Accelerometer,” *Microsystem Technologies*, vol. 24, no. 9, pp. 3851–3861, 2018.
- [J60] A. H. J. de Ruiter, J. R. Forbes, and M. Demjanenko, “On Constrained Kalman Filtering,” *AIAA Journal of Guidance, Control, and Dynamics*, vol. 41, no. 5, pp. 1206–1210, 2018.
- [J61] A. Walsh and J. R. Forbes, “Very Strictly Passive Controller Synthesis with Affine Parameter Dependence,” *IEEE Transactions on Automatic Control*, vol. 63, no. 5, pp. 1531–1537, 2018.
- [J62] R. J. Caverly and J. R. Forbes, “Flexible Planar Cable-Driven Parallel Manipulator Control: Maintaining Positive Cable Tensions,” *IEEE Transactions on Control Systems Technology*, vol. 26, no. 5, pp. 1874–1884, 2018.
- [J63] L. J. Bridgeman and J. R. Forbes, “Comparative Study of Input-Output Stability Results,” *IEEE Transactions on Automatic Control*, vol. 63, no. 2, pp. 463–476, 2018.
- [J64] S. S. J. R. Forbes, P. Seiler, and V. Gupta, “Conic-sector-based Analysis and Control Synthesis for Linear Parameter Varying Systems,” *IEEE Control Systems Letters*, vol. 2, no. 2, pp. 224–229, 2018.
- [J65] S. A. Chee and J. R. Forbes, “Discrete-Time Minmax Filtering Subject to a Norm-Constrained State Estimate,” *Automatica*, vol. 85, pp. 477–480, November 2017. DOI: 10.1016/j.automatica.2017.07.024.
- [J66] A. Walsh, J. R. Forbes, S. A. Chee, and J. J. Ryan, “Kalman Filter Based Unconstrained and Constrained Extremum-Seeking Guidance on  $SO(3)$ ,” *AIAA Journal of Guidance Control and Dynamics*, vol. 40, no. 9, pp. 2260–2271, 2017. **Special issue “The Kalman Filter and Its Aerospace Applications”**. DOI: 10.2514/1.G002635.
- [J67] A. H. J. de Ruiter and J. R. Forbes, “Continuous-Time Kalman Filtering on the Orthogonal Group  $O(n)$ ,” *International Journal of Robust and Nonlinear Control*, vol. 27, no. 17, pp. 3466–3487, 2017. DOI: 10.1002/rnc.3744.
- [J68] L. J. Bridgeman and J. R. Forbes, “Conic Bounds for Systems Subject to Unknown Delay,” *IEEE Transactions on Automatic Control*, vol. 62, no. 4, pp. 2006–2013, 2017.
- [J69] R. J. Caverly and J. R. Forbes, “Dynamic Modeling, Trajectory Optimization, and Control of a Flexible Kiteplane,” *IEEE Transactions on Control Systems Technology*, vol. 25, no. 4, pp. 1297–1306, 2017.
- [J70] A. Walsh and J. R. Forbes, “Analysis and Synthesis of Input Strictly Passive Gain-Scheduled Controllers,” *Journal of the Franklin Institute*, vol. 354, pp. 1285–1301, February 2017.
- [J71] D. E. Zlotnik and J. R. Forbes, “Nonlinear Estimator Design on the Special Orthogonal Group using Vector Measurements Directly,” *IEEE Transactions on Automatic Control*, vol. 62, no. 1, pp. 149–160, 2017.
- [J72] A. H. J. de Ruiter and J. R. Forbes, “Discrete-Time  $SO(n)$ -Constrained Kalman Filtering,” *AIAA Journal of Guidance Control and Dynamics*, vol. 40, no. 1, pp. 28–37, 2017.
- [J73] R. J. Caverly and J. R. Forbes, “Estimator Design for a Single Degree of Freedom Cable-Actuated System,” *Journal of the Franklin Institute*, vol. 353, no. 18, pp. 4845–4869, 2016.

- [J74] F. R. Hogan and J. R. Forbes, “Modeling of a Rolling Flexible Spherical Shell,” *ASME Journal of Applied Mechanics*, vol. 83, no. 9, pp. 091010(1–12), 2016.
- [J75] R. J. Caverly, C. Li, E. J. Chae, J. R. Forbes, and Y. L. Young, “Modeling and Control of Flow-Induced Vibrations of a Flexible Hydrofoil in Viscous Flow,” *Smart Materials and Structures*, vol. 25, no. 6, pp. 065007–065020, 2016.
- [J76] A. Walsh and J. R. Forbes, “A Very Strictly Passive Gain-Scheduled Controller: Theory and Experiments,” *IEEE/ASME Transactions on Mechatronics*, vol. 21, no. 6, pp. 2817–2826, 2016.
- [J77] D. E. Zlotnik and J. R. Forbes, “Exponential Convergence of a Nonlinear Attitude Estimator,” *Automatica*, vol. 72, pp. 11–18, October 2016.
- [J78] L. J. Bridgeman and J. R. Forbes, “The Extended Conic Sector Theorem,” *IEEE Transactions on Automatic Control*, vol. 61, pp. 1931–1937, July 2016.
- [J79] R. J. Caverly, D. E. Zlotnik, and J. R. Forbes, “Saturated Control of Flexible-Joint Manipulators Using a Hammerstein Strictly Positive Real Compensator,” *Robotica*, vol. 34, pp. 1367–1382, June 2016.
- [J80] F. R. Hogan and J. R. Forbes, “Trajectory Tracking, Estimation, and Control of a Pendulum-Driven Spherical Robot,” *AIAA Journal of Guidance Control and Dynamics*, vol. 39, no. 5, pp. 1119–1125, 2016.
- [J81] H. Sommer, J. R. Forbes, R. Siegwart, and P. Furgale, “Continuous-Time Estimation of Attitude Using B-Splines on Lie Groups,” *AIAA Journal of Guidance, Control, and Dynamics*, vol. 39, no. 2, pp. 242–261, 2016.
- [J82] F. R. Hogan and J. R. Forbes, “Modeling of a Rolling Flexible Circular Ring,” *ASME Journal of Applied Mechanics*, vol. 82, pp. 111003 (1–14), November 2015.
- [J83] A. Walsh and J. R. Forbes, “Modeling and Control of Flexible Telescoping Manipulators,” *IEEE Transactions on Robotics*, vol. 31, no. 4, pp. 936–947, 2015.
- [J84] A. H. J. de Ruiter and J. R. Forbes, “Generalized Euler Sequences Revisited,” *Journal of the Astronautical Sciences*, vol. 62, pp. 1–20, March 2015.
- [J85] J. R. Forbes, “Direction-Cosine-Matrix-Based Attitude Control Subject to Actuator Saturation,” *IET Control Theory and Applications*, vol. 9, no. 11, pp. 1653–1661, 2015.
- [J86] F. R. Hogan and J. R. Forbes, “Modeling of Spherical Robots Rolling on Generic Surfaces,” *Multi-body System Dynamics*, vol. 35, no. 1, pp. 91–109, 2015.
- [J87] L. J. Bridgeman and J. R. Forbes, “The Exterior Conic Sector Lemma,” *International Journal of Control*, vol. 88, no. 11, pp. 250–2263, 2015.
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- [J110] J. R. Forbes and C. J. Damaren, “Single-Link Flexible Manipulator Control Accommodating Passivity Violations: Theory and Experiments,” *IEEE Transactions on Control Systems Technology*, vol. 20, pp. 652–662, May 2012. **Nominated for “Outstanding Paper Award”.**
- [J111] J. R. Forbes and C. J. Damaren, “Design of Optimal Strictly Positive Real Controllers Using Numerical Optimization for the Control of Flexible Robotic Systems,” *Journal of the Franklin Institute*, vol. 348, no. 8, pp. 2191–2215, 2011.

- [J112] J. R. Forbes and C. J. Damaren, “Linear Time-Varying Passivity-Based Attitude Control Employing Magnetic and Mechanical Actuation,” *AIAA Journal of Guidance, Control, and Dynamics*, vol. 34, no. 5, pp. 1352–1362, 2011.
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- [J114] J. R. Forbes, T. D. Barfoot, and C. J. Damaren, “Dynamic Modeling and Stability Analysis of a Power-Generating Tumbleweed Rover,” *Multibody System Dynamics*, vol. 24, no. 4, pp. 413–439, 2010.
- [J115] J. R. Forbes and C. J. Damaren, “A Hybrid Passivity and Finite Gain Stability Theorem: Stability and Control of Systems Possessing Passivity Violations,” *IET Control Theory and Applications*, vol. 4, no. 9, pp. 1795–1806, 2010.
- [J116] J. R. Forbes and C. J. Damaren, “Design of Gain-Scheduled Strictly Positive Real Controllers Using Numerical Optimization for Flexible Robotic Systems,” *ASME Journal of Dynamic Systems, Measurement, and Control*, vol. 132, no. 3, p. 034503, 2010.
- [J117] J. R. Forbes and C. J. Damaren, “Geometric Approach to Spacecraft Attitude Control Using Magnetic and Mechanical Actuation,” *AIAA Journal of Guidance, Control, and Dynamics*, vol. 33, no. 2, pp. 590–595, 2010.

## Journal Publications Under Review

- [UR1] S. Dahdah and J. R. Forbes, “Uncertainty Modelling and Robust Observer Synthesis using the Koopman Operator,” *European Control Conference*, 2026.

## Refereed Conference Publications

Speaker indicated by ★.

- [RC1] Z. C. Guo★, J. R. Forbes, and T. Barfoot, “KILO-EKF: Koopman-Inspired Learned Observations Extended Kalman Filter,” *Conference on Intelligent Robots and Systems, Pittsburgh, PA, September 27 - October 1, 2026*.
- [RC2] M. Cohen★, V. Korotkine, and J. R. Forbes, “Observability and Consistency Analysis for Visual-Inertial Navigation with Anchored Feature Parameterizations,” *Conference on Intelligent Robots and Systems, Pittsburgh, PA, September 27 - October 1, 2026*.
- [RC3] S. Herschmann★, J. R. Forbes, C. Sloth, K. Zhang, and R. Skilton, “Continuous-Time Estimation of Deformation and Dynamic Parameters for Cosserat Rods under Free Vibration,” *IFAC World Congress, Busan, Republic of Korea, August 23-28, 2026*.
- [RC4] J. Eid★, A. Meagher, D. Rimorov, A. K. Bonala, R. Thike, and J. R. Forbes, “Power Hardware-in-the-loop Interfacing via H-infinity Model Matching,” *European Control Conference (ECC), Reykjavik, Iceland, July 7-10, 2026*.
- [RC5] D. Rimorov★ and J. R. Forbes, “Output Feedback Controller Synthesis with Finite Frequency Positive Realness Constraint,” *European Control Conference (ECC), Reykjavik, Iceland, July 7-10, 2026*.
- [RC6] T. E. Adams★, S. Dahdah, and J. R. Forbes, “dkpy: Robust Control with Structured Uncertainty in Python,” *European Control Conference (ECC), Reykjavik, Iceland, July 7-10, 2026*.
- [RC7] D. Rimorov★, J. R. Forbes, and R. Gagnon, “Measurement-Based Robust Stability Assessment of Interconnections Using Impedance Models,” *2026 IEEE Power and Energy Society General Meeting, Montreal, QC, July 19-23, 2026*.

- [RC8] S. Smocot\*, J. R. Forbes, and A. Sedal, “Observability-Informed Optimal Sensor Placement for Soft Robots,” *9th IEEE-RAS International Conference on Soft Robotics (RoboSoft 2026)*, Kanazawa, Japan, April 7-11, 2026.
- [RC9] S. Groen\*, D. Brenken, S. Smocot, J. R. Forbes, L. Mongeau, and A. Sedal, “Evaluation of Data-driven Kinematic Models for Autonomous Control of Continuum Robotic In-situ Bioprinters,” *9th IEEE-RAS International Conference on Soft Robotics (RoboSoft 2026)*, Kanazawa, Japan, April 7-11, 2026.
- [RC10] Y. Zhou\*, J. Kovecses, and J. R. Forbes, “Koopman Representation of Nonlinear Virtual Environments in Kinesthetic Haptic Systems,” *International Conference on Robotics and Automation*, Vienna, Austria, June 1-5, 2026.
- [RC11] S. Moalemi\* and J. R. Forbes, “Input-Output Stability of Gradient Descent: A Discrete-Time Passivity-Based Approach,” *American Control Conference*, Denver, CO, USA, July 8-10, 2025.
- [RC12] T. Hitchcox and J. R. Forbes, “An SE(3) Noise Model for Range-Azimuth-Elevation Sensors,” *Conference on Robots and Vision (CRV)*, Calgary, AB, May 26-29, 2025. Presented by A. Del Castillo Bernal\*.
- [RC13] Z. C. Guo\*, J. R. Forbes, and T. D. Barfoot, “Marginalizing and Conditioning Gaussians onto Linear Approximations of Smooth Manifolds with Applications in Robotics,” *International Conference on Robotics and Automation*, Atlanta, GA, May 19-23, 2025. **ICRA 2025 Best Paper Award.**
- [RC14] S. S. Ahmed\*, M. Shalaby, J. Le Ny, and J. R. Forbes, “Optimal Robot Formations: Balancing Range-Based Observability and User-Defined Configurations,” *Conference on Intelligent Robots and Systems*, Abu Dhabi, UAE, October 14-18, 2024.
- [RC15] S. S. Ahmed\*, M. Shalaby, C. Cossette, J. Le Ny, and J. R. Forbes, “Gaussian-Sum Filter for Range-based 3D Relative Pose Estimation in the Presence of Ambiguities,” *Conference on Control Technology and Applications*, Newcastle upon Tyne, UK, August 21-23, 2024.
- [RC16] S. Moalemi\* and J. R. Forbes, “Passivity-Based Gain-Scheduled Control with Scheduling Matrices,” *Conference on Control Technology and Applications*, Newcastle upon Tyne, UK, August 21-23, 2024.
- [RC17] J. Bramburger, S. Dahdah\*, and J. R. Forbes, “Synthesizing Control Laws from Data using Sum-of-Squares Optimization,” *Conference on Control Technology and Applications*, Newcastle upon Tyne, UK, August 21-23, 2024.
- [RC18] C. C. Cossette, M. Cohen, V. Korotkine, A. Del Castillo Bernal\*, M. A. Shalaby, and J. R. Forbes, “navlie: A Python Package for State Estimation on Lie Groups,” *Conference on Intelligent Robots and Systems*, Detroit, MI, October 1-5, 2023.
- [RC19] N. Pavlasek\*, C. C. Cossette, and J. R. Forbes, “Magnetic Navigation using Attitude-Invariant Magnetic Field Information for Loop Closure Detection,” *Conference on Intelligent Robots and Systems*, Detroit, MI, October 1-5, 2023.
- [RC20] K. Jung\*, T. Hitchcox, and J. R. Forbes, “Performance Evaluation of 3D Keypoint Detectors and Descriptors on Coloured Point Clouds in Subsea Environments,” *International Conference on Robotics and Automation*, London, UK, May 29 - June 2, 2023.
- [RC21] M. Shalaby\*, C. C. Cossette, J. R. Forbes, and J. Le Ny, “Calibration and Uncertainty Characterization for Ultra-Wideband Two-Way-Ranging Measurements,” *International Conference on Robotics and Automation*, London, UK, May 29 - June 2, 2023.
- [RC22] C. C. Cossette\*, M. Shalaby, D. Saussie, J. Le Ny, and J. R. Forbes, “Optimal Multi-robot Formations for Relative Pose Estimation Using Range Measurements,” *Conference on Intelligent Robots and Systems*, Kyoto, Japan, October 23-27, 2022.
- [RC23] M.-A. Lavoie\* and J. R. Forbes, “Map-Aided Train Navigation with IMU Measurements,” *Conference on Intelligent Robots and Systems*, Prague, Czech Republic (online), September 27 - October 1, 2021.

- [RC24] C. C. Cossette\*, M. Shalaby, D. Saussié, and J. R. Forbes, “Localization with Directional Coordinates,” *Conference on Intelligent Robots and Systems, Prague, Czech Republic (online), September 27 - October 1, 2021*.
- [RC25] N. Pavlasek\*, A. Walsh, and J. R. Forbes, “Invariant Extended Kalman Filtering Using Two Position Receivers for Extended Pose Estimation,” *International Conference on Robotics and Automation, Xi'an, China, May 30 - June 5, 2021*.
- [RC26] P. Laferriere\*, S. Laferriere, S. Dahdah, J. R. Forbes, and L. Paull, “Deep Koopman Representation for Control over Images (DKRCI),” *18th Conference on Robots and Vision, Burnaby, BC., May 26 - 28, 2021*.
- [RC27] T. Hitchcox\* and J. R. Forbes, “A Point Cloud Registration Pipeline using Gaussian Process Regression for Bathymetric SLAM,” *Conference on Intelligent Robots and Systems, Las Vegas, NV, October 25-29, 2020*.
- [RC28] D. Qian\*, G. Charland-Arcand, and J. R. Forbes, “TWOLATE: Total Registration of Point-Clouds Using a Weighted Optimal Linear Attitude and Translation Estimator,” *Conference on Control Technology and Applications, Montreal, QC, August 24-26, 2020*.
- [RC29] R. Fortune\*, C. A. Beltempo, and J. R. Forbes, “System Identification and  $\mathcal{H}_\infty$  Control of a Fatigue Structural Testing Rig,” *Conference on Control Technology and Applications, Montreal, QC, August 24-26, 2020*.
- [RC30] Q. Sun\*, J. Angeles, and J. R. Forbes, “Cartesian Elastodynamics Model of a Full-Mobility PKM with Flexible Links,” *New Trends in Mechanism and Machine Science. EuCoMeS 2020. Mechanisms and Machine Science*, vol. 89, pp. 305–313, 2020.
- [RC31] Q. Sun\*, J. Angeles, and J. R. Forbes, “Dynamics Modeling and Validation of a 3-PRR Planar Parallel Robot,” *Canadian Society for Mechanical Engineering International Congress, Charlottetown, PEI, June 21-24, 2020*.
- [RC32] K. Bergemin\*, D. Holden, S. Clavet, and J. R. Forbes, “DReCon: Data-Driven Responsive Control of Physics-Based Characters,” *SIGGRAPH Asia, Brisbane, Australia, November 18-21, 2019*.
- [RC33] D. Caverly, R. J. Caverly\*, and J. R. Forbes, “Periodic Tracking Control Using Gain-Scheduled Fourier Series-Based Internal Models,” *ASME Dynamic Systems and Control Conference, Park City, UT, October 9-11, 2019*.
- [RC34] M.-A. Lavoie\*, J. Arsenault, and J. R. Forbes, “An Invariant Extended  $\mathcal{H}_\infty$  Filter,” *Conference on Decision and Control, Nice, France, December 11-13, 2019*. Invited session 1 on “multi-sensor fusion techniques for state estimation in navigation”.
- [RC35] R. Fortune\*, C. A. Beltempo, and J. R. Forbes, “System Identification and Feedforward Control of a Fatigue Structural Testing Rig: The Single Actuator Case,” *21th IFAC Symposium on Automatic Control in Aerospace (ACA 2019), Cranfield UK, August 27-30, 2019*.
- [RC36] A. Walsh\*, J. Arsenault, and J. R. Forbes, “Invariant Sliding Window Filtering for Attitude and Bias Estimation,” *American Control Conference, Philadelphia, PA, USA, July 10-12, 2019*.
- [RC37] D. Qian\*, S. Rahman, and J. R. Forbes, “Relative Constrained SLAM for Robot Navigation,” *American Control Conference, Philadelphia, PA, USA, July 10-12, 2019*.
- [RC38] K. Lee\* and J. R. Forbes, “Synthesis of  $\mathcal{H}_\infty$ -Optimal Strictly Negative Imaginary Controllers,” *American Control Conference, Philadelphia, PA, USA, July 10-12, 2019*.
- [RC39] D. E. Zlotnik and J. R. Forbes\*, “Nonlinear Attitude and Bias Observer Design with a Gibbs-Inspired Cost Function Using Direct Vector Measurements,” *American Control Conference, Philadelphia, PA, USA, July 10-12, 2019*.
- [RC40] R. J. Caverly\* and J. R. Forbes, “ $\mathcal{H}_\infty$ -Optimal Strictly Positive Real Parallel Feedforward Control,” *American Control Conference, Philadelphia, PA, USA, July 10-12, 2019*.
- [RC41] Z. Yin\*, B. Belzile, J. Angeles, and J. R. Forbes, “Elastodynamics of a Parallel Schonflies-Motion Generator,” *2019 CCToMM M3 Symposium, Montreal, QC, May 16-17, 2019*.

- [RC42] R. Chiappinelli\*, M. Cohen, M. Doff-Sotta, M. Nahon, J. R. Forbes, and J. Apkarian, “Modeling and Control of a Passively-Coupled Tilt-Rotor Vertical Takeoff and Landing Aircraft,” *International Conference on Robotics and Automation, Montreal, QC, May 20-24, 2019*.
- [RC43] R. J. Caverly\* and J. R. Forbes, “ $\mathcal{H}_\infty$ -Optimal Parallel Feedforward Control using Minimum Gain,” *IEEE Conference on Decision and Control, Miami Beach, FL, December 17-19, 2018*.
- [RC44] R. J. Caverly and J. R. Forbes\*, “Linearly Combining Sensor Measurements Optimally to Enforce an SPR Transfer Matrix,” *IEEE Conference on Control Technology and Applications, Copenhagen, Denmark, August 21-24, 2018*.
- [RC45] K. Lee, A. Walsh, and J. R. Forbes\*, “Riemann Sphere Interpretation of the Passivity, Small Gain, and Conic Sector Theorems,” *IEEE Conference on Control Technology and Applications, Copenhagen, Denmark, August 21-24, 2018*.
- [RC46] R. Aucoin\*, S. A. Chee, and J. R. Forbes, “A Linear- and Linear-Matrix-Inequality-Constrained Extended Kalman Filter,” *American Control Conference, Milwaukee, WI, USA, June 27-29, 2018*.
- [RC47] L. J. Bridgeman\* and J. R. Forbes, “Iterative  $\mathcal{H}_2$ -Conic Controller Synthesis,” *American Control Conference, Milwaukee, WI, USA, June 27-29, 2018*.
- [RC48] Q. Sun\*, J. Angeles, and J. R. Forbes, “Elastostatics of a Full-Mobility PKM with Flexible Links,” *ROMANSY 22 - Robot Design, Dynamics and Control, Rennes, France, June 25-38, 2018*.
- [RC49] R. J. Caverly\* and J. R. Forbes, “Zero Shaping of Nonminimum Phase Aircraft Dynamics,” *AIAA Science and Technology Forum and Exposition 2018, Gaylord Palms, Kissimmee, FL, USA, January 9–12, 2018*.
- [RC50] J. R. Forbes\* and D. E. Zlotnik, “Sigma Point Kalman Filtering on Matrix Lie Groups,” *3rd conference on Geometric Science of Information (GSI 2017), Société Mathématique de France, Paris, France, November 7–9, 2017*. J.R. Forbes is an **invited speaker** as part of “Geometric Robotics” session.
- [RC51] E. Bering\*, L. Andersson, J. Chen, J. Cutler, K. Hara, D. Jackson, M. Lemmon, L. Pinsky, J. Sheehan, U. Siddiqui, R. Stoneback, P. Withers, R. Heelis, M. Moldwin, H. Reed, and J. Forbes, “Tri-metric Imaging of the Martian Ionosphere using a CubeSat Constellation,” *AIAA SPACE and Astronautics Forum and Exposition, Orlando, FL, USA, September 12-14, 2017*.
- [RC52] S. A. Chee\*, L. J. Bridgeman, and J. R. Forbes, “Norm- and Linear-Inequality-Constrained State Estimation: An LMI Approach,” *IEEE Conference on Control Technology and Applications, Kohala Coast, HI, USA, August 27–30, 2017*.
- [RC53] H. A. Godbole\*, R. J. Caverly, and J. R. Forbes, “Modelling of Flexible Cable-Driven Parallel Robots using a Rayleigh-Ritz Approach,” *Third International Conference on Cable-Driven Parallel Robots, Quebec City, QC, August 2-4, 2017*.
- [RC54] R. J. Caverly and J. R. Forbes\*, “Nyquist Interpretation of the Large Gain Theorem,” *20th IFAC World Congress, Toulouse, France, July 9–14, 2017*.
- [RC55] L. J. Bridgeman\* and J. R. Forbes, “Conic Controller Synthesis that Minimizes an Upper Bound on the Closed-Loop  $\mathcal{H}_2$ -Norm,” *American Control Conference, Seattle, WA, USA, May 24-26, 2017*.
- [RC56] R. J. Caverly\* and J. R. Forbes, “Regional Pole and Zero Placement with Static Output Feedback via the Modified Minimum Gain Lemma,” *American Control Conference, Seattle, WA, USA, May 24-26, 2017*.
- [RC57] A. Walsh\*, J. J. Ryan, and J. R. Forbes, “Constrained Extremum-Seeking Guidance Using a Constrained Kalman Filter,” *American Control Conference, Seattle, WA, USA, May 24-26, 2017*.
- [RC58] R. J. Caverly\*, J. R. Forbes, B. P. Danowsky, and P. M. Suh, “Gust-Load Alleviation of a Flexible Aircraft using a Disturbance Observer,” *AIAA Science and Technology Forum and Exposition 2017, Gaylord Texan, Grapevine, TX, USA, January 9–13, 2017*.

- [RC59] S. A. Chee\* and J. R. Forbes, “Norm-Constrained Forward-Backward Smoothing,” *AIAA Science and Technology Forum and Exposition 2017, Gaylord Texan, Grapevine, TX, USA, January 9–13, 2017*.
- [RC60] D. E. Zlotnik\* and J. R. Forbes, “Exteroceptive Measurement Filtering Embedded within an SO(3)-Based Attitude Estimator,” *55<sup>th</sup> Conference on Decision and Control, Las Vegas, NV, USA, December 12-14, 2016*.
- [RC61] R. J. Caverly\*, A. Girard, I. V. Kolmanovsky, and J. R. Forbes, “Nonlinear Dynamic Inversion of a Flexible Aircraft,” *20th IFAC Symposium on Automatic Control in Aerospace (ACA 2016), Sherbrooke, Quebec, Canada, August 21-15, 2016*.
- [RC62] S. A. Chee\*, J. R. Forbes, and D. S. Bernstein, “Gravity-Gradient-Stabilized Spacecraft Attitude Estimation Using Rate-Gyroscope Measurements,” *American Control Conference, Boston, MA, USA, July 6-8, 2016*.
- [RC63] D. E. Zlotnik\*, J. R. Forbes, and J. Aldrich, “Control Bandwidth Recovery of Flexible Pointing Systems,” *American Control Conference, Boston, MA, USA, July 6-8, 2016*.
- [RC64] R. J. Caverly\* and J. R. Forbes, “Robust Controller Design using the Large Gain Theorem: The Full-State Feedback Case,” *American Control Conference, Boston, MA, USA, July 6-8, 2016*. **Awarded “Best Paper” in session.**
- [RC65] R. J. Caverly\* and J. R. Forbes, “Flexible Kiteplane Modeling and Control with an Unsteady Aerodynamic Model,” *American Control Conference, Boston, MA, USA, July 6-8, 2016*.
- [RC66] E. A. Bering\*, L. S. Pinsky, L. Li, D. Jackson, J. Chen, H. Reed, M. Moldwin, J. Kasper, J. P. Sheehan, J. R. Forbes, T. Heine, A. Case, M. Stevens, and D. G. Sibeck, “MarsCAT: Mars Array of ionospheric Research Satellites using the CubeSat Ambipolar Thruster,” *AIAA Science and Technology Forum and Exposition 2016, San Diego, CA, USA, January 4–8, 2016*.
- [RC67] S. A. Chee\* and J. R. Forbes, “Norm-constrained Unscented Kalman Filter with Application to High Area-to-Mass Ratio Space-Debris Tracking,” *AIAA Science and Technology Forum and Exposition 2016, San Diego, CA, USA, January 4–8, 2016*.
- [RC68] F. R. Hogan\* and J. R. Forbes, “Trajectory Tracking of a Pendulum-Driven Spherical Robot,” *ASME 2015 Dynamic Systems and Control Conference, Columbus, OH, USA, October 28-30, 2015*.
- [RC69] F. R. Hogan\*, J. R. Forbes, and A. Walsh, “Dynamic Modeling of a Flexible Rolling Sphere,” *Proc. 11<sup>th</sup> International Conference on Multibody Systems, Nonlinear Dynamics, and Control, Boston, MA, USA, August 2-5, 2015*.
- [RC70] D. E. Zlotnik\* and J. R. Forbes, “A Nonlinear Attitude Estimator with Desirable Convergence Properties,” *European Control Conference, Linz, Austria, July 15-17, 2015*.
- [RC71] L. J. Bridgeman\*, M. Najih, and J. R. Forbes, “Strictly Positive Real and Conic System Syntheses Using Observers,” *American Control Conference, Chicago, IL, USA, July 1-3, 2015*.
- [RC72] A. Walsh\* and J. R. Forbes, “Control of a Planar Wind Energy Harvesting Kite with Flexible Cables,” *American Control Conference, Chicago, IL, USA, July 1-3, 2015*. Special session on Dynamics and Control of Airborne Wind Energy Systems.
- [RC73] R. J. Caverly\* and J. R. Forbes, “Maintaining Positive Cable Tensions During Operation of a Single Degree of Freedom Flexible Cable-Driven Parallel Manipulator,” *American Control Conference, Chicago, IL, USA, July 1-3, 2015*. **Awarded “Best Paper” in session.**
- [RC74] L. J. Bridgeman\* and J. R. Forbes, “Conic-Sector-Based Control in the Presence of Delay,” *12th IFAC Workshop on Time Delay Systems (TDS 2015), Ann Arbor, MI, USA, June 28-30, 2015*.
- [RC75] X. Shan\*, T. Zou, J. R. Forbes, and J. Angeles, “Design of Biaxial Navigation-grade MEMS Accelerometers,” *ASME 2014 International Mechanical Engineering Congress & Exposition, Montreal, QC, November 14-20, 2014*.
- [RC76] D. E. Zlotnik\* and J. R. Forbes, “Rotation-Matrix-Based Attitude Control Without Angular Velocity Measurements,” *American Control Conference, Portland, OR, USA, June 4-6, 2014*.

- [RC77] A. Weiss\*, F. Leve, M. Baldwin, J. R. Forbes, and I. Kolmanovsky, "Spacecraft Constrained Attitude Control using Positively Invariant Constraint Admissible Sets on  $SO(3) \times \mathbb{R}^3$ ," *American Control Conference, Portland, OR, USA, June 4-6, 2014*.
- [RC78] L. J. Bridgeman\* and J. R. Forbes, "The Minimum Gain Lemma," *American Control Conference, Portland, OR, USA, June 4-6, 2014*.
- [RC79] L. J. Bridgeman\*, R. J. Caverly, and J. R. Forbes, "Conic-Sector-Based Synthesis: Theory and Experiments," *American Control Conference, Portland, OR, USA, June 4-6, 2014*. **Awarded "Best Paper" in session.**
- [RC80] J. P. Stellini, J. R. Forbes\*, and C. J. Damaren, "Passivity-Based Adaptive Attitude Control with Gravity-Gradient Torque Compensation," *AIAA Guidance, Navigation, and Control Conference, Boston, MA, USA, August 19-22, 2013*.
- [RC81] L. J. Bridgeman\* and J. R. Forbes, "Conic-Sector-Based Control to Circumvent Passivity Violations," *American Control Conference, Washington, DC, USA, June 17-19, 2013*.
- [RC82] D. E. Zlotnik\* and J. R. Forbes, "Dynamic Modelling, Estimation, and Control for Precision Pointing of an Atmospheric Balloon Platform," *2013 CCToMM M3 Symposium, Montreal, QC, May 30-31, 2013*.
- [RC83] R. J. Caverly\*, D. E. Zlotnik, and J. R. Forbes, "Saturated Proportional Derivative Control of a Single-Link Flexible-Joint Manipulator," *2013 CCToMM M3 Symposium, Montreal, QC, May 30-31, 2013*.
- [RC84] T. Drews\*, P. Kry, J. R. Forbes, and C. Verbrugge, "Sequential Pose Estimation Using Linearized Rotation Matrices for Consumer Devices," *10<sup>th</sup> Conference on Computer and Robot Vision (CRV 2013), Regina, SK, May 28-31, 2013*. **T. Drews awarded "Best Student Paper" award.**
- [RC85] Y. Yang, I. Sharf, and J. R. Forbes\*, "Nonlinear Optimal Control of a Holonomic Airship," *AIAA Guidance, Navigation, and Control Conference, Minneapolis, MN, USA, August 13-16, 2012*. **Awarded "Best Presentation" in Control in Aerospace Robotics 1 session.**
- [RC86] J. R. Forbes\* and C. J. Damaren, "Synthesis of Optimal Finite Frequency Controllers for Flexible Robotic Manipulator Control," *AIAA Guidance, Navigation, and Control Conference, Portland, OR, USA, August 8-11, 2011*. **Awarded "Best Presentation" in Optimal Control session.**
- [RC87] E. J. Findlay\*, J. R. Forbes, H. H. T. Liu, A. H. J. de Ruiter, C. J. Damaren, and J. Lee, "Investigation of Active Vibration Suppression of a Flexible Satellite Using Magnetic Attitude Control," *AIAA Guidance, Navigation, and Control Conference, Portland, OR, USA, August 8-11, 2011*.
- [RC88] J. R. Forbes\*, "Extended Kalman Filter and Sigma Point Filter Approaches to Adaptive Filtering," *AIAA Guidance, Navigation, and Control Conference, Toronto, ON, August 2-5, 2010*. **AIAA GNC 2010 Graduate Student Paper Competition Finalist.**
- [RC89] J. R. Forbes\* and C. J. Damaren, "Linear Time-Varying Passivity-Based Attitude Control Employing Magnetic and Mechanical Actuation," *AIAA Guidance, Navigation, and Control Conference, Toronto, ON, August 2-5, 2010*.
- [RC90] J. R. Forbes\* and C. J. Damaren, "Passive Linear Time-Varying Systems: State-Space Realizations, Stability in Feedback, and Controller Synthesis," *American Control Conference, Baltimore, MD, USA, June 30 - July 2, 2010*.
- [RC91] J. R. Forbes\* and C. J. Damaren, "Robust Control of a Vibrating Beam Using the Hybrid Passivity and Finite Gain Stability Theorem," *AIAA Guidance, Navigation, and Control Conference, Chicago, IL, USA, August 10-13, 2009*, AIAA-2009-5787.
- [RC92] J. R. Forbes\* and C. J. Damaren, "Design of Gain-Scheduled SPR Controllers Using Numerical Optimization," *AIAA Guidance, Navigation, and Control Conference, Honolulu, HI, USA, August 18-21, 2008*, AIAA-2008-7312.

## Abstract Refereed Conference Publications

- [AC1] I. Loughheed\* and J. R. Forbes, “Range-Aided SLAM Initialization Exploiting Accurate Heading Information,” *OCEANS Conference and Exposition, Monterey CA, September 21-24, 2026*.
- [AC2] A. Meagher\*, J. Eid, D. Rimorov, and J. R. Forbes, “Performance Improvements to the Transmission Line Method used in Power Hardware-in-the-Loop Infrastructure,” *European Control Conference (ECC), Reykjavik, Iceland, July 7-10, 2026*.
- [AC3] S. S. Ahmed, N. Dahdah, M. A. Shalaby, C. C. Cossette, J. Le Ny, D. Saussié, and J. R. Forbes\*, “Multi-Robot Navigation using UWB in Infrastructure-Free Environments,” *40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA-X40), Rotterdam, NL, September 23-26, 2024*.
- [AC4] K. Jung, A. Del Castillo Bernal, T. Hitchcox, R. Wicks, and J. R. Forbes\*, “High-Resolution Mapping of Underwater Assets using a Laser Scanner,” *40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA-X40), Rotterdam, NL, September 23-26, 2024*.
- [AC5] S. Dahdah, J. R. Forbes\*, R. Vasudevan, and D. Bruder, “Identification and Control of Soft Robots Using the Koopman Operator,” *40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA-X40), Rotterdam, NL, September 23-26, 2024*.
- [AC6] L. Lortie\*, S. Dahdah, and J. R. Forbes, “Forward-Backward Extended DMD with an Asymptotic Stability Constraint,” *25th International Symposium on Mathematical Programming, Montreal, QC, July 21-16, 2024*.
- [AC7] S. Dahdah\* and J. R. Forbes, “Closed-loop Koopman Operator Approximation,” *25th International Symposium on Mathematical Programming, Montreal, QC, July 21-16, 2024*.
- [AC8] S. Moalemi\* and J. R. Forbes, “Input-Output Stability of First-Order Optimization Algorithms: A Passivity Approach,” *25th International Symposium on Mathematical Programming, Montreal, QC, July 21-16, 2024*.
- [AC9] J. Arsenault\*, M. Cunniff, E. Tulskey, and J. R. Forbes, “Spatial Roles in Hockey Special Teams,” *Cascadia Symposium on Statistics in Sports (CASSIS) 2022, September 24, 2022*.
- [AC10] J. R. Forbes\*, F. Ahmed, and M. Cohen, “Invariant Model Predictive Control of Quadrotor Vehicles,” *Robotics: Science and Systems, Online, July 12-16, 2021*. Presented in “Software Tools for Real-Time Optimal Control” workshop.
- [AC11] T. Hitchcox\* and J. R. Forbes, “Comparing Robust Cost Functions for Bathymetric Point Cloud Registration,” *2020 IEEE OES Autonomous Underwater Vehicle Symposium, St. John’s, NL, Sept. 30-Oct. 2, 2020*.
- [AC12] Q. Sun\*, J. Angeles, and J. R. Forbes, “Elastostatics of a Full-mobility PKM with Flexible Links,” *22<sup>nd</sup> CISM IFToMM Symposium on Robot Design, Dynamics and Control, Rennes, France, June 25-28, 2018*.
- [AC13] S. Papais\*, K. A. Carroll, and J. R. Forbes, “Spacecraft Planetary Navigation Using Radio Tracking and Accelerometer Measurements,” *18<sup>th</sup> CASI Astro Conference, Quebec City, QC, May 15-17, 2018*.
- [AC14] J. R. Forbes and A. H. de Ruiter, “Multibody System Dynamics: A Lagrangian Approach Using Any Constrained Attitude Parameterization,” *4th Joint International Conference on Multibody System Dynamics, Montreal, QC, May 29 - June 2, 2016*.
- [AC15] A. Walsh, D. E. Zlotnik, and J. R. Forbes\*, “Modelling of Spacecraft with  $N$  Reaction Wheels using Arbitrary Attitude Parameterizations,” *Advances in the Astronautical Sciences*, vol. 157, pp. 945–956, 2016. AAS Paper 16-138, Univelt, Inc., Escondido, CA. Presented at the 39th AAS Guidance, Navigation and Control Conference, Breckenridge, CO, USA, February 5 - 10, 2016.



- [AC16] D. E. Zlotnik\* and J. R. Forbes, “Effect of Pendulation on an  $SO(3)$ -Based Attitude Estimator for Precision Pointing of an Atmospheric Balloon-Borne Platform,” *2014 Academic High Altitude Conference, University of North Dakota, Grand Forks, ND, USA, June 25-27, 2014*.
- [AC17] N. K. Tran\*, D. E. Zlotnik, and J. R. Forbes, “Design of an Attitude Control System for a High-Altitude Balloon Payload,” *2013 Academic High Altitude Conference, Taylor University, Upland, IN, USA, June 26-28, 2013*.
- [AC18] N. K. Tran\*, X. He, D. E. Zlotnik, and J. R. Forbes, “Attitude Sensing and Control of a Stratospheric Balloon Platform,” *AIAA Balloon Systems Conference, Daytona Beach, FL, USA, March 25-28, 2013*.
- [AC19] A. Walsh\* and J. R. Forbes, “Dynamic Modelling and Control of Flexible Robotic Space Manipulators with Telescoping Booms,” *International Astronautical Congress, Naples, Italy, October 1 - 5, 2012, IAC-12, C1, 8, 7, x13683*.
- [AC20] A. Battiston\*, J. R. Forbes, B. Shmuel, and C. J. Damaren, “Power Consumption and Attitude Control in the Presence of Reaction Wheel Failure while using Reaction Wheels and Magnetic Torque Rods in Tandem,” *16<sup>th</sup> CASI Astro Conference, Quebec City, QC, April 24-26, 2012*.
- [AC21] E. J. Findlay\*, H. H. T. Liu, A. H. J. de Ruiter, C. J. Damaren, J. R. Forbes, and J. Lee, “Investigation of Satellite Flexibility due to Increased Drag Panel Area for Differential Drag Formation Flying,” *4<sup>th</sup> International Conference on Spacecraft Formation Flying Missions and Technologies, St. Hubert, QC, May 18-20, 2011*.
- [AC22] J. R. Forbes\* and C. J. Damaren, “Strictly Positive Real Controller Optimization for the Control of Flexible Manipulators,” *15<sup>th</sup> CASI Astro Conference, Toronto, ON, May 5-6, 2010, ASP-10-006*.

## Posters

- [PP1] I. Loughheed\* and J. R. Forbes, “LBL SLAM when Heading is Known, Part 2,” *2025 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 19, 2025*.
- [PP2] I. S. Requejo\* and J. R. Forbes, “Observability Analysis of Radar-Inertial Odometry,” *2025 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 19, 2025*.
- [PP3] M. Ghali\* and J. R. Forbes, “Control-Oriented Modelling of a Metal Burner,” *2025 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 19, 2025*.
- [PP4] I. Loughheed\* and J. R. Forbes, “LBL SLAM when Heading is Known, Part 1,” *2024 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 22, 2024*.
- [PP5] J. Yoo\* and J. R. Forbes, “Continuous-Time State Estimation,” *2024 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 22, 2024*.
- [PP6] A. Stirling\* and J. R. Forbes, “Machine Learning Approaches to UWB Noise Characterization,” *2024 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 22, 2024*.
- [PP7] S. Dahdah\* and J. R. Forbes, “Robust Koopman Observer Synthesis,” *Fourth Symposium on Machine Learning and Dynamical Systems, Toronto, ON, July 8-12, 2024*.
- [PP8] L. Lortie\* and J. R. Forbes, “Forward-Backward Extended DMD with an Asymptotic Stability Constraint,” *Fourth Symposium on Machine Learning and Dynamical Systems, Toronto, ON, July 8-12, 2024*.

- [PP9] S. S. Ahmed, M. A. Shalaby, J. L. Ny, and J. R. Forbes, "Optimal Robot Formations: Balancing Range-Based Observability and User-Defined Configurations," *ICAO Symposium, Montreal, QC, March 13, 2024*.
- [PP10] J. Eid\* and J. R. Forbes, "Implementing Antiwindup Compensation in Linear Control Systems," *2023 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 17, 2023*.
- [PP11] N. Dahdah\* and J. R. Forbes, "Learning Noise Characteristics of Object Detections for Multi-Object Tracking," *2023 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 17, 2023*.
- [PP12] S. Smocot\* and J. R. Forbes, "Building a Soft Robot," *2023 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 16, 2023*.
- [PP13] S. S. Ahmed\* and J. R. Forbes, "Formation and Attitude Control of Decentralized Multi-robot Systems," *2022 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 18, 2022*.
- [PP14] P. Decoste\* and J. R. Forbes, "Robot Navigation in Unknown Environments: Towards Topographical SLAM," *2021 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 17, 2021*.
- [PP15] J. Yun\* and J. R. Forbes, "Standardized Datasets for GNC Algorithm Development," *2021 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 17, 2021*.
- [PP16] S. Li\* and J. R. Forbes, "Dataset Collection for Benchmarking Robot Navigation Methods," *2021 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 17, 2021*.
- [PP17] A. Bianco\*, G. Ijpma, A.-M. Lauzon, and J. R. Forbes, "Automating Force Clamps to Measure Muscle Mechanics in Tissue Culture," *2020 Biomedical Engineering Society (BMES) Virtual Annual Meeting, October 14-17, 2020*.
- [PP18] N. Pavlasek\* and J. R. Forbes, "An Invariant Extended Kalman Filter for Navigation using Two Position Receivers and an Inertial Measurement Unit," *2020 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 26, 2020*.
- [PP19] D. Lisus\* and J. R. Forbes, "Learning Position and Orientation of a Robot Using Ultra-wideband Antenna Received Signal Strength," *2020 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 26, 2020*. Winner of the best SURE project, General and Multidisciplinary category.
- [PP20] A. Bianco\*, G. Ijpma, R. Zonis, A.-M. Lauzon, and J. R. Forbes, "Control of Nonlinear, Viscoelastic Tissue," *5TH Biological and Biomedical Engineering Symposium, McGill University, Montreal, QC, May 13-14, 2019*.
- [PP21] O. Nicoletti\* and J. R. Forbes, "Monocular Visual-Inertial Navigation," *2019 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 20, 2019*.
- [PP22] J. Jung\* and J. R. Forbes, "Autonomous Navigation of Robotic Systems Using the Invariant Sliding Window Filter," *2019 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 20, 2019*.
- [PP23] A. Al-Baali\* and J. R. Forbes, "Comparing Exteroceptive Sensors for Autonomous Underwater Vehicle Navigation," *2019 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 20, 2019*.
- [PP24] N. van der Laan, R. Fortune\*, and J. R. Forbes, "Fatigue Structural Testing Enhancement Research (FASTER)," *McGill Engineering Research Showcase (MERS), McGill University, Montreal, QC, September 28, 2018*.

- [PP25] J. Arsenault\* and J. R. Forbes, “Autonomous Underwater Navigation Using Visual-Inertial Odometry Aided INS,” *McGill Engineering Research Showcase (MERS), McGill University, Montreal, QC, September 27, 2018.*
- [PP26] S. Rahman\* and J. R. Forbes, “Simultaneous Localization and Mapping as a Probabilistic Constrained Optimization Problem,” *2018 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 9, 2018.*
- [PP27] N. van der Laan\*, R. Fortune, and J. R. Forbes, “Fatigue Structural Testing Enhancement Research (FASTER),” *2018 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 9, 2018.*
- [PP28] R. Habib\* and J. R. Forbes, “Variance-Constrained Control using Gaussian Process Models,” *2018 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 9, 2018.*
- [PP29] M. Cohen\* and J. R. Forbes, “Dynamic Modeling and Control of a Tilt-Rotor VTOL Aircraft,” *2018 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 9, 2018.*
- [PP30] R. Zonis\*, G. Ijpm, A.-M. Lauzon, and J. R. Forbes, “Force Feedback Control of Feedthrough-Exhibiting, Nonlinear, Viscoelastic, Biological Tissues,” *2018 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 9, 2018.*
- [PP31] J. Arsenault\* and J. R. Forbes, “GNC of an Autonomous Parafoil Deployed from a High-Altitude-Balloon,” *McGill Engineering Research Showcase (MERS), McGill University, Montreal, QC, September 25, 2017. Winner of the best MERS poster, Aerospace Engineering category.*
- [PP32] A. Walsh\* and J. R. Forbes, “Kalman-Filter-Based Extremum Seeking Guidance on SO(3),” *McGill Engineering Research Showcase (MERS), McGill University, Montreal, QC, September 25, 2017.*
- [PP33] D. Qian\* and J. R. Forbes, “Pose Estimation the OLATE Way,” *McGill Engineering Research Showcase (MERS), McGill University, Montreal, QC, September 25, 2017.*
- [PP34] J. Arsenault\* and J. R. Forbes, “High-Altitude-Balloon Parafoil Deployment and GNC,” *2017 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 10, 2017. Winner of the best SURE project, Aerospace Engineering category.*
- [PP35] R. Aucoin\* and J. R. Forbes, “LMI-Constrained State Estimation - EKF and UKF Approaches,” *2017 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 10, 2017.*
- [PP36] R. J. Caverly\* and J. R. Forbes, “Robust Controller Design Using the Large Gain Theorem,” *McGill Engineering Research Showcase (MERS), McGill University, Montreal, QC, October 4, 2016.*
- [PP37] H. A. Godbole\*, R. J. Caverly, and J. R. Forbes, “Passivity-Based Adaptive Control of a Cable-Actuated System,” *McGill Engineering Research Showcase (MERS), McGill University, Montreal, QC, October 4, 2016.*
- [PP38] C. Hart\* and J. R. Forbes, “Implementation of a Kalman Filter on a Google Tango Tablet,” *2016 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 11, 2016.*
- [PP39] A. Maalouly\* and J. R. Forbes, “Nyquist Interpretation of the Small Gain, Large Gain, and Passivity Theorems,” *2016 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 11, 2016.*
- [PP40] D. Caverly\* and J. R. Forbes, “Control of a Flexible-Link Manipulator with Zero Steady-State Tracking Error,” *2016 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 11, 2016. Winner of the best SURE project, Aerospace Engineering category.*

- [PP41] R. J. Caverly\* and J. R. Forbes, “Robust Controller Design Using the Large Gain Theorem,” *Seventh Biannual Meeting on Systems and Control Theory, Queen’s University, Kingston, ON, May 11-13, 2016*.
- [PP42] R. J. Caverly\* and J. R. Forbes, “Dynamic Modeling, Trajectory Optimization, and Control of a Flexible Kiteplane,” *Engineering Graduate Symposium, University of Michigan, Ann Arbor, MI, USA, October 30, 2015*.
- [PP43] L. J. Bridgeman\* and J. R. Forbes, “Conic-Sector-Based Control,” *McGill Engineering Research Showcase (MERS), McGill University, Montreal, QC, October 5, 2015*.
- [PP44] D. E. Zlotnik\* and J. R. Forbes, “Differential Geometric SLAM,” *International Conference on Robotics and Automation, Seattle, WA, USA, May 26-30, 2015*.
- [PP45] R. J. Caverly\*, E. J. Chae, J. R. Forbes, Y. L. Young, and D. S. Bernstein, “Modeling and Control of Flow-Induced Hydrofoil Vibrations,” *American Society of Naval Engineers, Arlington, VA, USA, March 4-5, 2015*. **2nd place in poster competition.**
- [PP46] L. J. Bridgeman\* and J. R. Forbes, “Conic-Sector-Based Control,” *Commonwealth Science Conference, Bangalore, India, November 25-28, 2014*.
- [PP47] F. R. Hogan\* and J. R. Forbes, “Dynamic Modelling of a Power-Generating Tumbleweed Rover,” *Commonwealth Science Conference, Bangalore, India, November 25-28, 2014*.
- [PP48] M. Najih\* and J. R. Forbes, “Using an Observer to Synthesize Optimal Strictly Positive Real and Conic Controllers,” *2013 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 15, 2013*.
- [PP49] F. R. Hogan\* and J. R. Forbes, “Dynamic Modelling and Stability Analysis of a Martian Tumbleweed Rover,” *2013 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 15, 2013*. **Winner of the best SURE project**, Aerospace Engineering category.
- [PP50] N. K. Tran\* and J. R. Forbes, “Attitude Control for a Stratospheric Balloon Platform,” *2013 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 15, 2013*.
- [PP51] R. J. Caverly\* and J. R. Forbes, “Dynamic Modelling and Passivity-Based Control of Cable-Actuated Systems,” *2013 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 15, 2013*.
- [PP52] D. E. Zlotnik\* and J. R. Forbes, “Flexible Spacecraft Modelling and Attitude Control in the Presence of Actuator Saturation,” *2012 McGill Summer Undergraduate Research in Engineering (SURE) Poster Presentation, Montreal, QC, August 16, 2012*.
- [PP53] N. K. Tran\* and J. R. Forbes, “Attitude Sensing and Control of a Stratosphere Balloon Platform,” *2012 Academic High Altitude Conference, Trevecca Nazarene University, Nashville, TN, USA, June 27-29, 2012*.
- [PP54] J. R. Forbes\* and C. J. Damaren, “Strictly Positive Real Controller Optimization for the Control of Flexible Manipulators,” *15<sup>th</sup> CASI Astro Conference, Toronto, ON, May 5-6, 2010, ASP-10-006*.

## Books

- [B1] A. H. J. de Ruiter, C. J. Damaren, and J. R. Forbes, *Spacecraft Dynamics and Control, An Introduction*. West Sussex, UK: John Wiley & Sons, Ltd., 2013.

## Theses

- [T1] J. R. Forbes, *Extensions of Input-Output Stability Theory and the Control of Aerospace Systems*. Ph.D. Thesis, University of Toronto, 2011.

- [T2] J. R. Forbes, *Design of Optimal Strictly Positive Real Controllers Using Numerical Optimization for the Control of Large Flexible Space Structures*. M.A.Sc. Thesis, University of Toronto, 2008.

## Industrial Patents

- [IP1] J. I. Saari, J. R. Forbes, N. Quitariano, and G. Roberts, “Chemical Sensor That Are Adaptable to Mobile Devices,” Patent Issued June 13, 2017. Canadian Patent Number: 869313.
- [IP2] J. I. Saari, N. Quitariano, J. R. Forbes, and G. Roberts, “Sensor Systems and Methods for Analyte detection,” Patent Issued March 24, 2016. Patent Number: US-20160084705.
- [IP3] P. Kambhampati, J. I. Saari, N. Quitariano, J. R. Forbes, and G. Roberts, “Systems for Detecting Target Chemicals and Methods for their Preparation and Use,” Patent Issued March 24, 2016. Patent Number: US-20160084810.

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## Presentations

Speaker indicated by “★”.

- [P1] J. R. Forbes★, “Learning Dynamic Systems with the Koopman Operator,” *Taylor Symposium on Applied AI, May 15, 2026*.
- [P2] J. R. Forbes★, “Robot Navigation using Ultra-Wideband Radio,” *University of Toronto, Robotics Institute, November 21, 2025*.
- [P3] J. R. Forbes★, “Gaussian Variational Inference for Batch Nonlinear State Estimation Applied to Robot Navigation,” *Department of Electrical Engineering, Eindhoven University of Technology, September 25, 2024*.
- [P4] J. R. Forbes★ and S. Dahdah, “Closed-loop Koopman Operator Approximation,” *Fourth Symposium on Machine Learning and Dynamical Systems, Toronto, ON, July 8-12, 2024*. **Invited Speaker.**
- [P5] S. S. Ahmed★, M. A. Shalaby, J. L. Ny, and J. R. Forbes, “Optimal Robot Formations: Balancing Range-Based Observability and User-Defined Configurations,” *ICAO Symposium, Montreal, QC, March 13, 2024*.
- [P6] J. R. Forbes, “SPRINT-Navs and Collabs,” *Soundwaves, The Sonardyne Podcast, March 1, 2024*.
- [P7] J. R. Forbes, “Multi-robot Relative Localization using Ultra-wideband Radio,” *RoboHacks, McGill University, Montreal, QC, February 17, 2024*.
- [P8] J. R. Forbes★ and T. Hitchcox, “Subsea Navigation and Mapping Using an Underwater Laser Scanner,” *ETH Zurich, Dep. of Mechanical and Process Engineering, Zurich, Switzerland, October 27, 2023*. **Invited Speaker.**
- [P9] J. R. Forbes★ and T. Hitchcox, “Subsea Navigation and Mapping Using an Underwater Laser Scanner,” *University of Klagenfurt, Department of Smart System Technologies, Klagenfurt, Austria, October 24, 2023*. **Invited Speaker.**
- [P10] J. R. Forbes★ and T. Hitchcox, “Subsea Navigation and Mapping Using an Underwater Laser Scanner,” *Australian Centre for Robotics, University of Sydney, Sydney, Australia, July 6, 2023*. **Invited Speaker** (online).
- [P11] J. R. Forbes★ and S. Dahdah, “Data-driven Koopman Operator Approximation,” *Koopman Operator Research Seminar, Dept. Mechanical Engineering, UC Santa Barbara, Santa Barbara, USA, April 21, 2023*. **Invited Speaker** (online).
- [P12] J. R. Forbes★, “Model Predictive Control of a Tandem-rotor Helicopter,” *State University of New York at Buffalo, Department of Mechanical and Aerospace Engineering, Amherst, NY, USA, February 23, 2023*. **Invited Speaker.**

- [P13] J. R. Forbes\* and T. Hitchcox, “Field Calibration of Optical Sensor Extrinsic in a Low-Observability Scenario,” *Endurance22 Science Workshop, Bremen, Germany, January 18-19, 2023*.
- [P14] T. Hitchcox\* and J. R. Forbes, “INS Drift Correction and Real-Time Subsea Point Cloud Colourization,” *Endurance22 Science Workshop, Bremen, Germany, January 18-19, 2023*.
- [P15] S. Dahdah\* and J. R. Forbes, “Learning Robot Dynamics with the Koopman Operator,” *IVADO’s Young Talent Development Program, Mila, October 21, 2022*. Best presentation in the “Algorithms of Tomorrow” session (\$1000 prize).
- [P16] J. R. Forbes\*, “Regularization Techniques in Koopman-based System Identification,” *Applied Mathematics Seminars, McGill University, September 12, 2022*.
- [P17] J. R. Forbes\*, “Model Predictive Control of a Tandem-rotor Helicopter,” *Dept. Electronics and Computer Science, University of Southampton, June 1, 2022*. **Invited Speaker**.
- [P18] J. R. Forbes\*, “Regularization Techniques in Koopman-based System Identification,” *Dept. Engineering Science, University of Oxford, May 30, 2022*. **Invited Speaker**.
- [P19] F. Ahmed and J. R. Forbes\*, “Model Predictive Control of a Tandem-rotor Helicopter,” *Optimization Days 2022, HEC Montreal, May 16-18, 2022*.
- [P20] S. Dahdah and J. R. Forbes\*, “System Norm Regularization Methods for Koopman Operator Approximation,” *Optimization Days 2022, HEC Montreal, May 16-18, 2022*.
- [P21] J. R. Forbes\*, “Regularization Techniques in Koopman-based System Identification,” *Ingenuity Labs Research Institute Lecture Series, Queen’s University, April 20, 2022*. **Invited Speaker**.
- [P22] J. R. Forbes\*, “MAME Podcase 1 - Professor James R. Forbes,” *McGill Association of Mechanical Engineers (MAME), August 11, 2020*.
- [P23] J. R. Forbes\*, “Gaussian Variational Inference for Batch Nonlinear State Estimation Applied to Robot Navigation,” *Meet a GERAD Researcher Seminar, Universite de Montreal, May 6, 2020*.
- [P24] J. R. Forbes\*, “Gaussian Variational Inference for Batch Nonlinear State Estimation Applied to Robot Navigation,” *Robot Learning Seminar Series, Mila, March 20, 2020*. **Invited Speaker**.
- [P25] J. R. Forbes\*, “Wind-Velocity Estimation for Enhanced UAV Path Planning and Control,” *Ingenuity Labs Research Institute Lecture Series, Queen’s University, February 12, 2020*. **Invited Speaker**.
- [P26] A. Bianco\*, G. Ijpma, R. Zonis, A.-M. Lauzon, and J. R. Forbes, “Control of Nonlinear, Viscoelastic Tissue,” *30th Annual RI-MUHC Research Day on Monday, June 3rd, 2019*.
- [P27] T. Hitchcox\* and J. R. Forbes, “Multiscale Gaussian Process Regression for Feature Enhancement in Laser-based Bathymetric SLAM,” *Optimization Days 2019, HEC Montreal, May 13-15, 2019*.
- [P28] K. Lee\* and J. R. Forbes, “Synthesis of Optimal Yet Robust Controllers for Negative Imaginary Systems,” *Optimization Days 2019, HEC Montreal, May 13-15, 2019*.
- [P29] J. R. Forbes\* and R. Fortune, “System Identification and Optimal Control of a Fatigue Testing Rig for Aircraft,” *Optimization Days 2019, HEC Montreal, May 13-15, 2019*.
- [P30] J. R. Forbes, “Pose Estimation via the Weighted Optimal Linear Translational Attitude Estimation (WOLATE) Algorithm,” *Lund University, Lund, Sweden, August 20, 2018*. **Invited Speaker**.
- [P31] J. R. Forbes, “ $\mathcal{H}_\infty$ -Optimal Parallel Feedforward Control,” *8th Biennial Meeting on Systems and Control Theory, University of Toronto, Toronto, ON, May 7-9, 2018*.
- [P32] J. R. Forbes, “Extremum-Seeking Guidance on the Special Orthogonal Group,” *Aerial Robotics International Research Symposium, Centre for Aerial Robotics Research and Education (CARRE), Toronto, ON, June 21-22, 2017*. **Invited Speaker**.
- [P33] J. R. Forbes, “Guidance on  $SO(3)$  Using a Kalman Filter,” *Tutorial Session “Vision-based Navigation and Perception for Space Systems”, American Control Conference, Seattle, WA, USA, May 24-26, 2017*. **Invited Speaker**.
- [P34] R. J. Caverly\* and J. R. Forbes, “Controller Design for Regional Pole and Zero Placement using Linear Matrix Inequalities and the Modified Minimum Gain Lemma,” *Optimization Days 2017, HEC Montreal, May 8-10, 2017*.

- [P35] A. Walsh and J. R. Forbes\*, “Extremum-Seeking Guidance on  $SO(3)$  Using a Kalman Filter,” *Optimization Days 2017, HEC Montreal, May 8-10, 2017*.
- [P36] D. Qian and J. R. Forbes\*, “Exploiting landmark Constraints for Mobile robot Localization and Mapping,” *Optimization Days 2017, HEC Montreal, May 8-10, 2017*.
- [P37] J. R. Forbes, “So, you want to be a Professor, eh?,” *Mech Talks, MAME, McGill University, Montreal, QC, April 4, 2017. Invited Speaker.*
- [P38] R. J. Caverly and J. R. Forbes, “Robust Controller Design using the Large Gain Theorem,” *GERAD Student Day, HEC Montreal, QC, April 11, 2017*.
- [P39] D. Qian\* and J. Forbes, “Pose Estimation Using a Novel Computer Vision Algorithm,” *GAMES Graduate Research Day, McGill University, Montreal, QC, March 8, 2017. Awarded Second Place in the Oral Presentation Section.*
- [P40] R. J. Caverly\* and J. Forbes, “Dynamics, Estimation, and Control Lab Research Overview,” *McGill Association of Mechanical Engineers (MAME) Mech Lab Tour, McGill University, Montreal, QC, January 23, 2017*.
- [P41] J. R. Forbes, “Observer Design on  $SO(3)$ ,” *G  RAD, Montreal, QC, December 2, 2016. Invited Speaker.*
- [P42] J. R. Forbes, “State Estimation for Aerial Vehicles — Probabilistic and Deterministic Approaches,” *KittyHawk, Mountain View, CA, USA, November 17, 2016. Invited Speaker.*
- [P43] J. R. Forbes, “Robust Control of Flexible Spacecraft,” *Arizona State University, Department of Mechanical and Aerospace Engineering, Tempe, AZ, USA, October 28, 2016. Invited Speaker.*
- [P44] R. J. Caverly\* and J. Forbes, “Robust Controller Design via the Large Gain Theorem,” *Third Annual CIM-REPARTI Research Showcase, Centre for Intelligent Machines (CIM), McGill University, Montreal, QC, October 19, 2017*.
- [P45] J. R. Forbes, “Dynamic Modeling, Estimation, and Control of Cable-Actuated Robots,” *William Maxwell Reed Seminar Series Invited Speaker, University of Kentucky, Department of Mechanical and Aerospace Engineering, Lexington, KY, USA, October 14, 2016*.
- [P46] R. J. Caverly\* and J. Forbes, “The Large Gain Theorem and its Applications,” *Informal Systems Seminar (ISS), McGill University, Montreal, QC, October 7, 2017*.
- [P47] J. R. Forbes, “Control and Estimation on the Special Orthogonal Group of Rigid-Body Rotations . . . and Other Adventures,” *State University of New York at Buffalo, Department of Mechanical and Aerospace Engineering, Amherst, NY, USA, September 29, 2016. Invited Speaker.*
- [P48] J. R. Forbes, “How to be Successful in Your NSERC Fellowship Application — My Thoughts on the Process,” *How to be Successful in Your NSERC Fellowship Application Workshop, SKILLSETS, McGill University, Montreal, QC, August 31, 2015, August 29, 2016. Invited Speaker.*
- [P49] J. R. Forbes, “Observer Design on  $SO(3)$  - Fast Convergence and Practical Considerations,” *7th Biennial Meeting on Systems and Control Theory, Queen’s University, Kingston, ON, May 11-13, 2016*.
- [P50] J. R. Forbes, “Dynamic Modeling, Estimation, and Control of Cable-Actuated Robots,” *Max Planck Institute for Biological Cybernetics, T  bingen, Germany, May 10, 2016. Invited Speaker.*
- [P51] J. R. Forbes, “Estimation on  $SO(3)$ ,” *Mini-symposium on Geometry in Mechanics in Honour of Paul Zsombor-Murray’s 80<sup>th</sup> Birthday, McGill University, Montreal, QC, April 22, 2016. Invited Speaker.*
- [P52] L. J. Bridgeman\* and J. R. Forbes, “Methods Revitalizing Input-Output Stability Analysis,” *2016 Mechanical Engineering Graduate Research Day, McGill University, Montreal, QC, March 10, 2016. Co-winner for best presentation.*
- [P53] R. J. Caverly\* and J. R. Forbes, “Robust Controller Design using the Large Gain Theorem,” *2016 Mechanical Engineering Graduate Research Day, McGill University, Montreal, QC, March 10, 2016*.

- [P54] R. J. Caverly\* and J. Forbes, “Dynamics, Estimation, and Control Lab Research Overview,” *McGill Association of Mechanical Engineers (MAME) Mech Lab Tour, McGill University, Montreal, QC, January 19, 2016.*
- [P55] J. R. Forbes, “Becoming an Engineering Academic: My Road to McGill,” *MIAE Guest Speaker Series Event, McGill University, Montreal, QC, October 6, 2015. Invited Speaker.*
- [P56] J. R. Forbes, “Control and Estimation on the Special Orthogonal Group of Rigid-Body Rotations ... and Other Adventures,” *Texas A & M University, Department of Aerospace Engineering, Collage Station, TX, USA, September 24, 2015. Invited Speaker.*
- [P57] D. E. Zlotnik\* and J. R. Forbes, “A Nonlinear Attitude Estimator with Desirable convergence Properties,” *First Michigan Aerospace Control Conference (MACC), University of Michigan, Ann Arbor, MI, USA, May 18, 2015.*
- [P58] S. A. Chee\* and J. R. Forbes, “Modeling and Estimation of Flexible Space Debris,” *First Michigan Aerospace Control Conference (MACC), University of Michigan, Ann Arbor, MI, USA, May 18, 2015.*
- [P59] R. J. Caverly\* and J. R. Forbes, “Modeling and Control of a Wind Energy Harvesting Flexible Kiteplane,” *First Michigan Aerospace Control Conference (MACC), University of Michigan, Ann Arbor, MI, USA, May 18, 2015.*
- [P60] J. R. Forbes, “The Conic Sector Theorem: The Past, The Present, The Future,” *University of Notre Dame Dept. of Electrical Engineering Systems Seminar, Notre Dame, IN, USA, April 10, 2015. Invited Speaker.*
- [P61] J. R. Forbes, “Estimation and Control of Aerial Robotic Systems,” *IEEE Robotics and Automation Society, Southeastern Michigan Chapter, Lawrence Technological University, Southfield, MI, USA, March 21, 2015. Invited Speaker.*
- [P62] J. R. Forbes, “Flexible Aircraft Modeling and Gain-Scheduled Control,” *Air Force Research Laboratory, Wright-Patterson Air Force Base, Dayton, OH, USA, February 17, 2015. Invited Speaker.*
- [P63] J. R. Forbes, “Attitude Estimation on  $SO(3)$ ,” *Air Force Research Laboratory, Kirtland Air Force Base, Albuquerque, NM, USA, January 26, 2015. Invited Speaker.*
- [P64] J. R. Forbes, “Recent Developments in Attitude Estimation and the Control of Flexible Aircraft,” *NASA Armstrong Flight Research Center, Edwards Air Force Base, CA, USA, December 18, 2014. Invited Speaker.*
- [P65] J. R. Forbes, “Attitude Estimation and Control of Spacecraft and Large Flexible Space Structures,” *Jet Propulsion Laboratory, Pasadena, CA, USA, December 16, 2014. Invited Speaker.*
- [P66] J. R. Forbes, “Control and Estimation of Aerospace Systems,” *Presentation given to the Department of Aerospace Engineering, University of Michigan, Ann Arbor, MI, USA, December 12, 2014.*
- [P67] J. R. Forbes, “Control and Estimation of Aerospace Systems,” *Presentation given to the Department of Aerospace Engineering’s Industrial Advisory Board, University of Michigan, Ann Arbor, MI, USA, October 24, 2014.*
- [P68] J. R. Forbes, “Borne To Point — Dynamic Modeling, Estimation, and Control of Balloon-Borne Payloads,” *Aerospace Control and Guidance Systems Committee Meeting #114, Cleveland, OH, USA, October 15-17, 2014. Invited Speaker, awarded “Best Presentation” Award.*
- [P69] J. R. Forbes, “Dynamic Modeling, Estimation, and Control of Aerospace Systems,” *Universities Space Research Association, Madison, WI, USA, October 9, 2014. Invited Speaker.*
- [P70] J. R. Forbes, “Guidance, Navigation, and Control Techniques for Balloon-Borne Payloads,” *NASA Glenn Research Center, Cleveland, OH, USA, April 7, 2014. Invited Speaker.*
- [P71] J. R. Forbes, “High-Altitude Balloon Navigation, Guidance, and Control,” *Google, Mountain View, CA, USA, January 31, 2014. Invited Speaker.*
- [P72] J. R. Forbes, “Dynamic Modelling, Estimation, and Control of High-Altitude Balloons, Spherical Rovers, and Spacecraft,” *Jet Propulsion Laboratory, Pasadena, CA, USA, January 30, 2014. Invited Speaker.*



- [P73] J. R. Forbes, "Spacecraft Attitude Estimation and Control: Norm-Constrained Kalman Filtering and Actuator Saturation Prevention," *The Aerospace Corporation, Los Angeles, CA, USA, January 28, 2014. Invited Speaker.*
- [P74] J. R. Forbes, "Gain-Scheduled Control Within a Passive Systems Framework," *NASA Dryden Flight Research Center, Edwards Air Force Base, CA, USA, January 27, 2014. Invited Speaker.*
- [P75] J. R. Forbes, "Gain-Scheduled Control Within a Passive Systems Framework," *NASA Ames Research Center, Mountain View, CA, USA, December 11, 2013. Invited Speaker.*
- [P76] J. R. Forbes, "Guidance, Navigation, and Control Techniques for Balloon-Borne Payloads," *NASA Glenn Research Center, Cleveland, OH, USA, December 3, 2013. Invited Speaker.*
- [P77] J. R. Forbes, "Practicable Rotation-Matrix-Based Attitude Control," *Control Systems Seminar, University of Michigan, Ann Arbor, MI, USA, November 8, 2013.*
- [P78] J. R. Forbes, "Attitude Sensing and Control of Small Balloon-Borne Payloads," *NASA Glenn Research Center, Cleveland, OH, USA, October 22, 2013. Invited Speaker.*
- [P79] L. J. Bridgeman\* and J. R. Forbes, "Conic-Sector-Based Control to Circumvent Passivity Violations," *Department of Electrical Engineering, University of Notre Dame, South Bend, ID, USA, May 23, 2013.*
- [P80] L. J. Bridgeman\* and J. R. Forbes, "Conic-Sector-Based Control to Circumvent Passivity Violations," *2013 Optimization Days, Montreal, QC, Canada, May 6-8, 2013.*
- [P81] J. R. Forbes, "Attitude Control and Estimation of Aerospace Systems," *AEROSP 585 - Aerospace Engineering Seminar, Department of Aerospace Engineering, University of Michigan, Ann Arbor, MI, USA, March 12, 2013. Invited Speaker.*
- [P82] J. R. Forbes, "Control and Estimation of Aerospace and Robotic Systems," *Presented to Lockheed Martin personnel, McGill University, Montreal, QC, June 4, 2012.*
- [P83] J. R. Forbes, "Control of Aerospace Systems," *McGill-CSA Space Day, McGill University, Montreal, QC, March 1, 2012. Invited Speaker.*
- [P84] J. R. Forbes, "Becoming an Engineering Academic: My Road to McGill," *McGill Association of Mechanical Engineers, McGill University, Montreal, QC, February 28, 2012. Invited Speaker.*
- [P85] J. R. Forbes, "Passivity-Based Control Using the Rotation Matrix (and Other Adventures)," *Informal Systems Seminar, Centre for Intelligent Machines, McGill University, Montreal, QC, December 15, 2011.*
- [P86] J. R. Forbes, "Linear, Nonlinear, and Optimal Passivity-Based Control," *Dynamics and Control Seminar, Sibley School of Mechanical and Aerospace Engineering, Cornell University, Ithaca, NY, USA, December 5, 2011. Invited Speaker.*
- [P87] J. R. Forbes, "Conic Sectors and the Conic Sector Theorem: The Past, the Present, and the Future," *Informal Systems Seminar (ISS), McGill University, Montreal, QC, November 18, 2011.*
- [P88] J. R. Forbes, "Advances in Passivity-Based Control for Aerospace Systems," *AEROSP 585 - Aerospace Engineering Seminar, Department of Aerospace Engineering, University of Michigan, Ann Arbor, MI, USA, September 15, 2011. Invited Speaker.*
- [P89] J. R. Forbes, "Advances in Passivity-Based Control for Aerospace Systems," *presented to the UTAIS Advisory Board, UTIAS, Toronto, ON, May 12, 2011.*
- [P90] J. R. Forbes\* and C. J. Damaren, "Flexible Manipulator Control Accommodating Passivity Violations," *Space Vision and Advanced Robotics Workshop 2011, MDA Space Missions, Brampton, ON, June 8, 2011.*
- [P91] J. R. Forbes, "Extended Kalman Filter and Sigma Point Filter Approaches to Adaptive Filtering," *Space Vision and Advanced Robotics Workshop 2010, MDA Space Missions, Brampton, ON, June 9, 2010.*

- [P92] J. R. Forbes\* and C. J. Damaren, “Flexible Manipulator Control Accommodating Passivity Violations: Theory and Experiments,” *Meeting on Systems and Control Theory, Queen’s University, Kingston, ON, May 13-15, 2010*.
- [P93] J. R. Forbes, “Robust Control of a Vibrating Beam Using the Hybrid Passivity and Finite Gain Stability Theorem,” *International Astronautical Congress, Daejeon, Republic of Korea, October 16-19, 2009*.
- [P94] J. R. Forbes\* and C. J. Damaren, “Design of Gain-Scheduled SPR Controllers Using Numerical Optimization to Control Flexible Space Systems,” *Space Vision and Advanced Robotics Workshop 2009, MDA Space Missions, Brampton, ON, June 3, 2009*.

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## Public Media Citations and Interviews

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|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| November 9, 2024 | Nico Vincent interviewed by Matthieu Dugal on the Radio Canada program “Moteur de recherche”. Mr. Vincent explained DECAR and Voyis’ contribution to the surveying of the Endurance shipwreck. |
| October 17, 2024 | Interview with La Presse regarding the Endurance shipwreck.                                                                                                                                    |
| October 15, 2024 | Interview with CityNews Montreal regarding the Endurance shipwreck.                                                                                                                            |
| October 12, 2024 | Interview with the Montreal Gazette regarding the Endurance shipwreck.                                                                                                                         |
| October 11, 2024 | Interviewed by Jill Bennett for the Jill Bennett Show on CKNW (Vancouver).                                                                                                                     |

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## Awards, Recognition, and Nominations

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|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| April 2, 2026    | Winner of the <b>Donald L. Mordell EUS Teaching Excellence Award</b>                                                                                                                                                                                                                                                                                                    |
| May 22, 2025     | International Conference on Robotics and Automation (ICRA) <b>best paper award</b> for the paper:<br>Zi Cong Guo, James R. Forbes, Timothy D. Barfoot, “Marginalizing and Conditioning Gaussians onto Linear Approximations of Smooth Manifolds with Applications in Robotics,” International Conference on Robotics and Automation, Atlanta, GA, May 19-23, 2025       |
| April 24, 2024   | Winner of the <b>Samuel and Ida Fromson Outstanding Teaching Award in Engineering</b>                                                                                                                                                                                                                                                                                   |
| June 2021        | International Conference on Robotics and Automation (ICRA) <b>best student paper award finalist</b> (1 of 4 finalists) for the paper:<br>M. Shalaby, C. C. Cossette, J. Le Ny, and J. R. Forbes, “Cascaded Filtering Using the Sigma Point Transformation,” <i>IEEE Robotics and Automation Letters</i> , vol. 6, no. 3, pp. 4758-4765, 2021. Jointly accepted to ICRA. |
| April 2020       | Winner of the <b>Carrie M. Derick Award for Graduate Supervision and Teaching</b>                                                                                                                                                                                                                                                                                       |
| January 25, 2020 | The paper “DReCon: Data-Driven responsive Control of Physics-Based Characters” co-authored by Prof. Forbes, MSc student K. Bergemin, and collaborators from Ubisoft La Forge featured on “Two Minute Papers”:<br><a href="https://youtu.be/o_DhNqHazKY">https://youtu.be/o_DhNqHazKY</a>                                                                                |
| May 31, 2017     | Winner of the <b>Engineering Class of 1944 Outstanding Teaching Award</b>                                                                                                                                                                                                                                                                                               |

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|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2017, 2014       | Nominated for the ACGSC Dave Ward Memorial Lecture Award                                                                                                                                           |
| April 7, 2016    | Winner of the <b>McGill Association of Mechanical Engineers (MAME) “Professor of the Year”</b>                                                                                                     |
| March 12, 2015   | Nominated for the University of Michigan’s <b>Golden Apple Award</b> that honours instructors who aim to “. . . not only to disseminate knowledge but inspire and engage students in its pursuit”. |
| 2012             | Nominated for the Governor General’s Gold Medal for the highest academic standing at the graduate level.                                                                                           |
| November 3, 2011 | <b>G.N. Patterson Award</b> , given to the most outstanding Ph.D. graduate of UTIAS.                                                                                                               |
| 2011 - 2013      | NSERC Post Doctoral Fellowship (awarded but declined)                                                                                                                                              |
| August 3, 2010   | AIAA Graduate Student Paper Competition Finalist                                                                                                                                                   |
| 2008 - 2011      | Ontario Graduate Scholarship                                                                                                                                                                       |
| 2009             | Canadian Space Agency Travel Award                                                                                                                                                                 |
| 2009             | Bronze “T” Award, Varsity Lacrosse, University of Toronto                                                                                                                                          |
| 2007 - 2008      | Ontario Graduate Scholarship in Science and Technology                                                                                                                                             |
| 2005-2006        | University of Waterloo Faculty of Engineering Upper-Year Scholarship                                                                                                                               |

## Conference Awards

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|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| October 21, 2022   | “Best presentation” in the “algorithms of tomorrow” session at IVADO’s Young Talent Development event. Work co-authored by S. Dahdah and J.R. Forbes. Presentation given by S. Dahdah. |
| July 6-8, 2016     | “Best presentation” in Robust Control I session, American Control Conference, Boston, MA, USA.                                                                                         |
| June 1-3, 2015     | “Best presentation” in Control Applications II session, American Control Conference, Chicago, IL, USA.                                                                                 |
| June 4-6, 2014     | “Best presentation” in Robust Control I session, American Control Conference, Portland, OR, USA.                                                                                       |
| May 28-31, 2013    | “Best student paper”, 10 <sup>th</sup> Conference on Computer and Robot Vision (CRV 2013), Regina, SK, Canada.                                                                         |
| August 13-16, 2012 | “Best presentation” in Control in Aerospace Robotics 1 session, AIAA Guidance, Navigation, and Control Conference, Minneapolis, MN, USA.                                               |
| August 8-11, 2011  | “Best presentation” in Optimal Control session, AIAA Guidance, Navigation, and Control Conference, Portland, OR, USA.                                                                  |

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## Research Grants

### McGill University

#### Research Grants Awarded

- NSERC Alliance with Provectus Robotics Solutions (PI) \$515,013 CAD + industry in-kind, 2025-2027  
*Multi-modal Calibration and Multi-robot Navigation in Unstructured Environments*

- United Arab Emirates University (Co-PI) \$300,000 CAD, 2026-2027  
*Reinforcement Learning and Energy-Efficient TinyML Approach for VTOL UAVs*
- Canadian Space Agency (Co-PI) \$73,500 CAD, 2024-2026  
*Advancing Deep Space Propulsion Technology Using High-Altitude Balloon Platforms*
- NSERC CREATE ADVENTOR Training Program (Co-PI) \$1,515,000 CAD, 2024-2030  
*Advanced Engineering and Training in Next-Generation Mobile Robotics for Human Spaces*
- Mitacs Accelerate with ARA Robotique, Inc. \$36,000 CAD, 2024  
*Pose Estimation of Objects of Interest by Aerial Multirotor Drone*
- NSERC Alliance/Mitacs with Voyis and Ontario Power Generation (PI) \$513,444 CAD + industry in-kind, 2023-2028  
*Multi-Sensor Navigation and Scene Reconstruction in Subsea Environments*
- NSERC Alliance with Hydro-Quebec's Research institute (IREQ) and OPAL-RT Technologies Inc. (PI) \$116,536 CAD + industry in-kind, 2023-2025  
*Development of Robust Power-Hardware-In-the-Loop Interface Algorithms*
- **NSERC Discovery Grant** \$250,000 CAD (\$50,000/year), 2023 - 2028  
*Enhanced Robustness and Performance of Navigation Solutions for Robotic Vehicles*
- DND/NSERC Discovery Grant Supplements \$120,000 CAD (\$40,000/year), 2023 - 2026  
*Enhanced Robustness and Performance of Navigation Solutions for Robotic Vehicles*
- FRQNT NOVA (Co-PI) \$225,000 CAD, 2023-2026  
*Multi-UAV Active Simultaneous Localization and Mapping in Unknown Indoor Environments*
- Mitacs Accelerate with Haply \$45,000 CAD, 2022-2024  
*Improving Dynamic Range by Using Rich Sensing and Gain-scheduled Control for Improved Haptic Feedback*
- NSERC Alliance Mission with ARA Robotique, Inc. (PI) \$441,872 CAD + industry in-kind, 2022-2024  
*Infrastructure Inspection Using a Team of Unmanned Aerial Vehicles*
- Mitacs Accelerate with SBQuantum + IVADO Supplement \$45,000 CAD, 2021  
*Simultaneous localization and mapping using a magnetic quantum sensor*
- NSERC Alliance with Denso \$89,366 CAD + industry cash and in-kind, 2021-2024  
*Automotive Visual-inertial Navigation*
- IVADO projets de recherche fondamentale (co-PI) \$224,598 CAD, 2020-2022  
*Differentiable perception, graphics, and optimization for weakly supervised 3D perception*
- Mitacs Accelerate with Mecademic, Inc. \$90,000 CAD, 2021-2023  
*Data-Driven Control of an Ultracompact Industrial Robot*
- Mitacs Accelerate with NGC Aerospace, Inc. \$60,000 CAD, 2020-2022  
*Guidance and Control of a Tandem-rotor Helicopter Landing on a Moving Platform*
- Mitacs Accelerate with ARA Robotique, Inc. \$30,000 CAD, 2020-2021  
*Tightly coupled visual-inertial-LIDAR SLAM for real time application*
- CIFAR AI Catalyst Grant (co-PI) \$100,000 CAD (\$50,000/year), 2020-2022  
*Modeling Embodied Agents with Koopman Embeddings*
- IDeAS (co-PI) \$1,583,025 + in kind, 2020-2023  
*Cooperative Network of Autonomous Unmanned Vehicles Protection, Trustworthiness, and Resilient Recovery Subject to Faults and Cyber Attacks*
- Mitacs Accelerate with ARA Robotique, Inc. \$15,000 CAD, 2020  
*Wind Field Estimation, Exploitation and Guidance for Unmanned Aerial Vehicles*
- Mitacs Accelerate with Thales \$30,000 CAD, 2020  
*Constrained Batch Estimation for Train Positioning with Inertial Sensors*
- Research Contract with the Carolina Hurricanes Hockey Club \$155,400 CAD, 2020-2023

- Data-Driven Decision Making Using Player and Puck Tracking Data*
- Research Contract with Mecademic \$51,800 CAD, 2020-2022
  - Research Contract with Mecademic \$28,000 CAD, 2019-2020
  - Mitacs Accelerate with ARA Robotique, Inc. (co-PI) \$30,000 CAD, 2019-2020
- Tightly-coupled Visual-Inertial-LiDAR SLAM*
- Mitacs Accelerate with Mecademic, Inc. \$53,333 CAD, 2019
- High-Performance Control of an Ultra-Compact Industrial Robot Arm*
- Mitacs Accelerate with Reaction Dynamics \$15,000 CAD, 2019
- Research and Experimental Testing of Liquid-Injection Thrust Vector Control Actuator (LITVC)*
- Mitacs Accelerate with Thales \$30,000 CAD, 2019
- Constrained Kalman Filtering for Train Position Estimation*
- NSERC Engage with Reaction Dynamics \$25,000 CAD, 2019
- Control, Navigation and Guidance Concept Studies [Pre-Phase A] for a Venture Class Orbital Launch Vehicle (5 - 150kgs)*
- Mitacs Accelerate with ARA Robotique, Inc. \$30,000 CAD, 2019
- Guidance and Control of Hybrid Vertical Takeoff and Landing (VTOL) Unmanned Aerial Vehicles (UAVs) with Thrust Vectoring Capabilities*
- Canadian Foundation for Innovation JELF \$125,000 CAD, 2019-2022
- State Estimation Enabling Autonomous Operation of Robotic Vehicles*
- Mitacs Accelerate with Ubisoft, Inc. \$30,000 CAD, 2018
- RLCapture: A deep reinforcement learning based control strategy for switching between motion capture inspired controllers*
- Mitacs Accelerate with GlobVision, Inc. \$16,850 CAD, 2018-2019
- Detection of spacecraft maneuvers based on publicly available data*
- Mitacs Accelerate with ARA Robotique, Inc. \$30,000 CAD, 2018-2019
- Weighted Linear Point-cloud Registration for Scan-matching in GPS-denied environments*
- NSERC Collaborative Research and Development (CRD) with 2G Robotics, Inc. (PI) \$377,668 CAD (\$94,417/year) + industry cash and in-kind, 2018-2022
- Enhancing Subsea Navigation Capabilities*
- Research Contract with National Research Council of Canada \$90,000 CAD, 2017-2019
- FASTER - Fatigue Structural Testing Enhancement Research*
- FRQNT Projets de recherche en équipe (co-PI) \$205,740 CAD, 2018-2021
- Cooperative Navigation and Control of Mobile Robots Using UWB Radio*
- NSERC Collaborative Research and Development (CRD) Grant with CMLabs, Inc. (co-PI) \$450,000 CAD, 2018-2021
- Efficient Modelling and Simulation of Mechanical Systems*
- NSERC Connect \$3,525 CAD, 2017
- Facilitation meeting with 2G Robotics*
- Mitacs Accelerate with ARA Robotique, Inc. \$30,000 CAD, 2017-2018
- 3D RFM SLAM for UAVs Subject to Severe Vibration*
- NSERC Engage with ARA Robotique, Inc. \$25,000 CAD + \$10,000 CAD CRIAC Sup., 2017
- GPS/INS Performance Enhancement for UAVs*
- NSERC Engage with GlobVision, Inc. \$25,000 CAD + \$10,000 CAD CRIAC Supplement, 2016
- Advanced Data Analysis to Enhance Space Situational Awareness*
- **NSERC Discovery Grant** \$198,000 CAD (\$33,000/year), 2016 - 2023
- Enhanced Performance, Stability, and Practicability of Attitude and Position Estimators for Robotic Vehicles*
- CFI Leaders Opportunity Fund \$125,000 CAD, 2013 - 2018

*Practicable Control, Estimation, and Sensing Methods for Aerospace and Robotics Systems*

- Canadian Space Agency (co-PI) \$24,000 CAD, 2013 - 2014  
*ALTAIR: Flights for Precision Calibration for Dark Energy, Microwave Astronomy, and Atmospheric Physics*  
The amount \$24,000 CAD represents my share of a larger grant PI'ed by Profs. J. Albert (UVic).
- **FRQNT Établissement de Nouveaux Chercheurs** \$81,586 CAD, 2012 - 2014  
*Control and Estimation of Dynamic Systems on the Special Orthogonal Group*
- **NSERC Discovery Grant** \$115,000 CAD (\$23,000/year), 2012 - 2017  
*Development of Novel Input-Output Control Schemes for Mechanical, Aerospace, and Robotic Systems*
- McGill University, Start-up Funds \$80,000 CAD, 2011

## The University of Michigan

- NSF EAGER \$100,002 USD (\$131,647.63 CAD), 2015-2017  
*DG-SLAM: Differential Geometric Simultaneous Localization and Mapping*
- NASA AERO Institute \$79,000.00 USD (\$102,700 CAD), 2016  
*Constrained Peak-Seeking Control via Constrained Kalman Filtering*
- **NASA Early Career Faculty Award** \$599,146 USD (\$788,746 CAD), 2015-2018  
*Dynamics and Control of Tensegrity Space Manipulators*
- NASA STTR grant with Systems Technology Inc. (STI) (co-PI) \$55,000 USD (\$72,405 CAD), 2015-2016  
*Gust Load Estimation and Rejection with Application to Robust Flight Control Design for HALE Aircraft*
- NASA Jet Propulsion Lab grant \$15,000 USD (\$19,747 CAD), 2015  
*MuTip Pointing Control for Model-Free High-bandwidth Imaging Precision*
- NASA Jet Propulsion Lab grant \$15,647 USD (\$20,599 CAD), 2015  
*Dynamic Modelling of Cubesats for Mission Feasibility Simulator*
- NASA Langley Research Center grant (PI with two other co-PIs) \$62,858 USD (\$82,748 CAD), 2013-2015  
*Heliogyro Blade High-altitude Environment Characterization at the University of Michigan*
- University of Michigan, Start-up Funds \$565,000 USD, 2013

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## Professional Activities and Affiliations

### Journal Editorial Boards

|                                     |                                                                |
|-------------------------------------|----------------------------------------------------------------|
| September 1, 2024                   | Senior Editor<br>International Journal of Robotics Research    |
| February 23, 2023 - August 31, 2024 | Associate Editor<br>International Journal of Robotics Research |

### Conference Editorial Boards and Program Committees

|             |                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------|
| 2022 - 2025 | IEEE Robotics and Automation Society Editorial Board<br>Associate Editor for IROS (area of Mechanisms, Design and Control) |
|-------------|----------------------------------------------------------------------------------------------------------------------------|

|                |                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------|
| 2013 - Present | IEEE Control Systems Society Conference Editorial Board<br>Associate Editor for ACC and CDC                                        |
| 2024 - 2025    | Program Committee for ACC 2025                                                                                                     |
| 2017 - 2018    | Program Committee of the 15th Conference on Computer and Robot Vision (CRV), Toronto, Canada, May 9-11, 2018                       |
| 2011 - 2012    | Technical Program Committee<br>7th Annual CIGRE Conference on Power Systems Conference<br>Montreal, QC<br>September 24 - 26, 2012. |

### Technical Committees and Society Membership

|                |                                                                                              |
|----------------|----------------------------------------------------------------------------------------------|
| 2024 - Present | IEEE Technical Committee on Robust and Complex Systems                                       |
| 2013 - Present | IEEE Technical Committee on Aerospace Controls                                               |
| 2012 - Present | IEEE Robotics and Automation Society                                                         |
| 2011 - Present | IEEE Control Systems Society                                                                 |
| 2014 - 2020    | Aerospace Control and Guidance Systems Committee (ACGSC)<br>ACGSC representative to the IEEE |

### Conference Sessions Chaired

|                       |                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| July 8-10, 2025       | American Control Conference<br>Sessions: Control Interpretations of Optimization Algorithms                                             |
| May 19 - 23, 2025     | International Conference on Robotics and Automation<br>Session: Sensor Fusion 1                                                         |
| August 21-23, 2024    | Conference on Control Technology and Applications<br>Session: Nonlinear Systems                                                         |
| May 13-17, 2024       | International Conference on Robotics and Automation<br>Session: Sensor Fusion II                                                        |
| May 29 - June 2, 2023 | International Conference on Robotics and Automation<br>Session: Marine and Field Robotics                                               |
| May 16-18, 2022       | Optimization Days<br>Session: Robotics                                                                                                  |
| May 30 - June 5, 2021 | International Conference on Robotics and Automation<br>Session: Multiple and Distributed Systems VIII                                   |
| August 24-26, 2020    | Control Theory and Applications<br>Session: Aerospace Applications                                                                      |
| August 21-24, 2018    | Control Theory and Applications<br>Session: Linear Systems                                                                              |
| June 27-29, 2018      | American Control Conference<br>Sessions: Optimal Control II, Optimal Control V                                                          |
| May 24-26, 2017       | American Control Conference<br>Sessions: Linear Systems I, Optimal Control II, Vision-based Navigation and Perception for Space Systems |
| July 6-8, 2016        | American Control Conference<br>Session: Robust Control I                                                                                |

|                        |                                                                                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| May 30 - June 2, 2016  | 4th Joint International Conference on Multibody System Dynamics (IMSB)<br>Session: MOD1 - Modelling, Formalism and Theoretical Methods |
| July 1-3, 2015         | American Control Conference<br>Session: Control Applications II                                                                        |
| August 19-22, 2013     | AIAA Guidance Navigation and Control Conference<br>Session: Flexible and Articulated Spacecraft Dynamics and Control                   |
| June 17-19, 2013       | American Control Conference<br>Session: Stability of Nonlinear Systems IV                                                              |
| August 13-16, 2012     | AIAA Guidance Navigation and Control Conference<br>Session: Robotic Systems in Aerospace                                               |
| June 30 - July 2, 2010 | American Control Conference<br>Session: Linear Systems 2                                                                               |

### Conference, Workshops, and Special Sessions Organized and Co-organized

|                           |                                                                                                                                                                                                     |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| August 11-15, 2025        | 2025 Montreal Robotics Summer School<br>Expert speaker on “State Estimation for Robotics”                                                                                                           |
| July 8-10, 2025           | American Control Conference Special Session, Special Session Co-organizer<br>Session Title: <i>Control Interpretations of Optimization Algorithms</i>                                               |
| June 21-25, 2025          | RSS Workshop on Equivariant Systems<br>Panelist                                                                                                                                                     |
| June 12-13, 2025          | 2025 ADVENTOR Training Week<br>Leader of <i>Planning, Navigation, and Mapping</i> training module. Member of <i>Machine Learning and Control</i> module. Lecture on “State Estimation for Robotics” |
| February 12 - May 7, 2025 | NASA / GNC TDT Robust Control Analysis and Design Course<br>Presented in collaboration with Peter Seiler, Julian Theis, James Forbes                                                                |
| May 7-9, 2025             | Meeting on System and Control Theory, Conference Co-organizer<br>Primary local organizer                                                                                                            |
| July 10-12, 2024          | American Control Conference Special Session, Special Session Co-organizer<br>Session Title: <i>Recent Systems and Control Research in Canada</i>                                                    |
| June 11-14, 2024          | 2024 Montreal Robotics Summer School<br>Expert speaker on “State Estimation for Robotics”                                                                                                           |
| May 3-5, 2023             | Meeting on System and Control Theory, Conference Co-organizer                                                                                                                                       |
| August 22-25, 2022        | 2022 Montreal Robotics Summer School<br>Expert speaker on “State Estimation for Robotics”                                                                                                           |
| May 16, 2022              | 2022 ELATE Teaching and Learning Conference<br>Host of “Flipped Tutorial” breakout session                                                                                                          |
| April 27-29, 2020         | Meeting on System and Control Theory, Conference Co-organizer<br>Note: this event was ultimately cancelled due to the 2020 COVID-19 pandemic.                                                       |
| June 4-6, 2014            | American Control Conference Special Session, Special Session Co-organizer<br>Session Title: <i>SO(3)-Based Attitude Control and Estimation</i>                                                      |
| March 13-16, 2012         | Bellairs Workshop on Computer Animation, Workshop Co-organizer<br>Session Title: <i>Pseudospectral Optimal Control for Character Animation</i>                                                      |



## Reviews

Many reviews have been completed for the following funding organizations, journals, and conferences. The year, and the number of reviews completed in that year, is indicated in parenthesis in the form “20XX-Y” where “20XX” indicates the year and “Y” indicates the number of reviews in that year.

### Reviews for Funding Organizations

NSERC Discovery Grant Program (2013-2, 2017-1, 2019-1)  
NSERC Alliance Program (2022-2)  
Dutch Research Council (NWO), Domain Applied and Engineering Sciences (AES) (2023-1)  
National Geographic (2017-1)  
New Frontiers in Research Fund (2022-1)  
IVADO (2017-1)  
Mitacs (2017-1, 2019-1, 2020-1, 2023-1, 2026-1)  
Canadian Foundation for Innovation (2016-1, 2026-1)  
NASA Missouri Space Grant/EPSCoR (2014-1)

### Journal Paper Reviews

Acta Astronautica (2012-2, 2020-1)  
Aerospace Systems (2024-1)  
AIAA Journal (2015-1)  
AIAA Journal of Guidance, Control, and Dynamics (2010-2, 2011-1, 2014-5, 2015-2, 2016-1, 2017-6, 2018-3, 2021-3, 2022-1, 2023-2, 2024-2). Recognized as an “**excellent reviewer**” in 2013, 2014, 2015, 2016, 2017, and 2021.  
AIAA Journal of Spacecraft and Rockets (2013-1, 2014-1, 2017-1)  
AIAA Journal of Aerospace Information Systems (2017-1, 2018-1)  
American Mathematical Monthly (2016-2)  
Annual Reviews in Control, Robotics, and Autonomous Systems (2020-1)  
ASME Journal of Computational and Nonlinear Dynamics (2011-1, 2012-1, 2016-3, 2023-1)  
ASME Journal of Dynamics, Measurement, and Control (2015-1, 2017-1, 2023-1)  
Journal of Field Robotics (2014-2, 2019-1)  
Journal of Control Science and Engineering (2017-1)  
Journal of Multibody System Dynamics (2022-1)  
Automatica (2013-1, 2016-2, 2017-1, 2022-1, 2024-2)  
Chinese Journal of Aeronautics (2013-1)  
European Journal of Control (2013-1)  
Hypothesis (2009-1, 2010-1)  
IEEE Transactions on Automatic Control (2014-1, 2016-2, 2017-3, 2018-1, 2021-1, 2022-4, 2023-1, 2025-1)  
IEEE Control Systems Letters (2017-2, 2018-1, 2020-1, 2022-4, 2024-1, 2026-1)  
IEEE Transactions on Aerospace and Electronic Systems (2012-1, 2013-1, 2016-1, 2017-1, 2021-1, 2024-2, 2026-2)  
IEEE Transactions on Control Systems Technology (2014-1, 2015-1, 2016-1, 2019-1, 2020-1, 2021-2, 2022-1, 2023-1, 2024-1, 2026-1)  
IEEE Transactions on Control of Network Systems (2019-1)  
IEEE Transactions on Circuits and Systems (2011-1)  
IEEE Transactions on Field Robotics (2026-1)

IEEE Transactions on Robotics (2014-1, 2015-2, 2016-2, 2019-1, 2020-1, 2021-1, 2023-1, 2024-2)  
 IEEE Robotics and Automation Letters (2017-1, 2019-1, 2020-1, 2021-2, 2022-3, 2023-2, 2024-1, 2026-1)  
 IEEE Transactions on Industrial Electronics (2021-1)  
 IEEE Control Systems Magazine (2010-2, 2023-1)  
 International Journal of Robust and Nonlinear Control (2012-1, 2013-1)  
 International Journal of Control (2012-1, 2021-1, 2022-1)  
 International Journal of Aerospace Engineering (2014-1)  
 International Journal of Robotics Research (2016-1, 2026-1)  
 Journal of the Franklin Institute (2015-1, 2016-1)  
 Journal of Small Satellites (2012-1)  
 Optimization and Engineering (2016-1)  
 Mechanism and Machine Theory (2012-1)  
 Mathematics of Control, Signals, and Systems (2012-1)  
 Proc. Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering (2014-1, 2015-1)  
 Royal Aeronautical Society (2014-1)  
 Systems and Control Letters (2018-1)  
 The Aeronautical Journal (2014-1)  
 Transactions of the Institute of Measurement and Control (2012-2)  
 Transactions of the Canadian Society for Mechanical Engineering (2012-1, 2014-1, 2016-1)

### Conference Paper Reviews

AIAA SciTech (2015-2, 2018-2)  
 AIAA Guidance, Navigation, and Control Conference (2009-1, 2010-1, 2011-3, 2013-3)  
 APCA International Conference on Automatic Control and Soft Computing (2022-1)  
 CCToMM Symposium on Mechanisms, Machines, and Mechatronics (2013-2)  
 European Control Conference (ECC) (2025-1)  
 Geometric Science of Information (2017-2)  
 IEEE Canadian Conference on Electrical & Computer Engineering (2012-4)  
 IEEE Conference on Robotics and Automation (ICRA) (2010-1, 2011-2, 2014-1, 2016-1, 2019-2, 2022-2, 2023-1, 2024-4, 2025-3)  
 IEEE Conference on Intelligent Robots and Systems (IROS) (2011-1, 2012-1, 2014-2, 2015-1, 2016-1, 2017-1, 2022-1, 2023-1, 2024-3, 2026-1)  
 IEEE American Control Conference (ACC) (2010-1, 2011-1, 2013-1, 2014-1, 2016-2, 2017-4, 2019-1, 2021-5, 2022-3, 2023-1, 2024-4, 2025-2)  
 IEEE Conference on Decision and Control (CDC) (2010-1, 2012-1, 2013-2, 2014-3, 2017-4, 2018-1, 2021-2, 2022-4, 2023-2, 2024-5, 2026-3)  
 IEEE Conference on Control Technology and Applications (CCTA) (2017-2, 2018-1, 2024-1)  
 IEEE Multi-Conference on Systems and Control (MSC) (2013-1)  
 IFAC World Congress (2011-1, 2013-1, 2023-2)  
 IFAC Mathematical Theory of Networks and Systems (2024-1)  
 IFAC Symposium on Nonlinear Control Systems (2016-1)  
 IFAC Symposium on Automatic Control in Aerospace (2016-4)  
 International Conference on Space Robotics (iSpaRo) (2026-1)  
 Robotics: Systems and Science (2023-4, 2025-3, 2026-4).  
 SIGGRAPH Asia (2023-1).

## Book Reviews for Publishers

Book review (2013-1, 2019-2, 2022-2) for Cambridge University Press.

Book review (2014-1) for Wiley.

## Archival Book Reviews

J. R. Forbes, “Review of ‘ $L_2$ -Gain and Passivity Techniques in Nonlinear Control’ by A. van der Schaft”, *IEEE Control Systems Magazine*, vol. 37, no. 6, December 2017.

J. R. Forbes, “Review of ‘Fundamentals of Spacecraft Attitude Determination and Control’ by F. Landis Markley and John L. Crassidis”, *IEEE Control Systems Magazine*, vol. 36, no. 4, August 2015.

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## Administrative University Activities

### McGill University

#### Committee Memberships within the Dept. of Mechanical Engineering, McGill University

|                                                                            |                                                           |
|----------------------------------------------------------------------------|-----------------------------------------------------------|
| March 26, 2024 - June 30, 2024 &<br>March 25, 2020 - July 31, 2021         | Member of Search Committee for Mech. Engineering<br>Chair |
| September 1, 2021 - Present                                                | Department Academic Committee                             |
| September 1, 2021 - Present                                                | Tenure and Promotion Committee                            |
| September 1, 2021 - Present                                                | U3/U4, U0/U1, U2, Academic Advisor                        |
| September 1, 2020 - Present                                                | Chair of Curriculum Review Committee                      |
| September 1, 2017 - August 31, 2020                                        | Member of Curriculum Review Committee                     |
| September 1, 2018 - April 30, 2019                                         | Mechanical Engineering Search Committee                   |
| August 1, 2015 - August 31, 2018                                           | Graduate Admissions and Scholarships Committee            |
| May 16, 2011 - August 31, 2013                                             | Systems, Measurement, and Controls Committee              |
| September 1, 2016 - Present, &<br>January 1, 2012 - August 31, 2013        | Undergraduate Teaching Labs Committee                     |
| August 1, 2015 - August 31, 2017, &<br>September 1, 2012 - August 31, 2013 | Mechanical Engineering Colloquium organizer               |
| September 1, 2012 - August 31, 2013                                        | MECH 609 - Seminar organizer                              |

#### Additional Service within the Dept. of Mechanical Engineering, McGill University

|                                     |                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------|
| September 1, 2015 - August 31, 2016 | Spearheaded “midterm teaching evaluation” program with Teaching and Learning Services. |
| January 1, 2016 - August 31, 2016   | Spearheaded the Mechanical Engineering Undergraduate Exit Survey.                      |
| August 1, 2015 - August 31, 2016    | Spearheaded the creation of the <code>matlab</code> Manual for Undergraduate Students. |

#### Committee Memberships within the Faculty of Engineering, McGill University

|                           |                                       |
|---------------------------|---------------------------------------|
| April 25, 2025 - Present  | Student Standing and Awards Committee |
| October 9, 2024 - Present | Mathematics Course Review Committee   |

|                                                                 |                                                                                                        |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| September 12, 2024 - Present, & 2016 - 2018                     | Committee on Teaching and Learning                                                                     |
| November 10, 2023 - Present                                     | Graduate Engineering Equity Committee (GEEC) Faculty Advisor                                           |
| April 2024                                                      | Faculty-level committee for ranking of William Dawson Scholar and James McGill Professor Applications. |
| March 2018 - May 2018                                           | Search Committee for the Chair of the Department of Electrical and Computer Engineering                |
| 2016 - 2018                                                     | “Leadership Centre” Development Committee                                                              |
| August 1, 2015 - Present, & September 1, 2012 - August 31, 2013 | MIAE Steering Committee                                                                                |
| September 1, 2012 - August 31, 2013                             | MIAE Undergraduate and Graduate Curriculum Committee                                                   |
| August 1, 2015 - Present, & May 16, 2011 - August 31, 2013      | McGill Institute for Aerospace Engineering (MIAE)                                                      |

### **Other Service Items at McGill University and the Larger Academic Community**

|                                     |                                                                                                                            |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 2025                                | College of Graduate Mentorship                                                                                             |
| 2024                                | As a “leading researcher in robotics” I completed a survey “what is a robot”.                                              |
| 2022                                | Reviewer of 6 dossiers submitted for the GERAD Ph.D. Thesis Excellence Award                                               |
| 2021                                | Reviewer of 3 dossiers submitted for the RSS Pioneers Workshop                                                             |
| 2021                                | Reviewer of 6 dossiers submitted for the CAGS Award for Outstanding Graduate Mentorship                                    |
| 2021                                | Reviewer of 9 dossiers submitted for the Carrie M. Derick Award for Graduate Supervision and Teaching                      |
| September 1, 2016 - August 31, 2018 | Faculty advisor for the McGill Space Group (MSG)                                                                           |
| September 1, 2016 - August 1, 2017  | Interim faculty advisor for McGill Robotics                                                                                |
| March 26, 2017                      | Robohacks Judge                                                                                                            |
| August 29, 2016                     | Speaker at “How to be successful in your NSERC fellowship application workshop”, SKILLSETS, Teaching and Learning Services |
| August 31, 2015                     |                                                                                                                            |

### **Other Committees**

|                |                                   |
|----------------|-----------------------------------|
| 2018 - Present | Vanier Science Advisory Committee |
|----------------|-----------------------------------|

### **University of Michigan**

#### **Committee Memberships within the Dept. of Aerospace Engineering, University of Michigan**

|                                   |                                                  |
|-----------------------------------|--------------------------------------------------|
| September 1, 2013 - July 31, 2015 | Flight Dynamics and Control Curriculum Committee |
|-----------------------------------|--------------------------------------------------|

## Committee Memberships within the College of Engineering, University of Michigan

|                                   |                                                  |
|-----------------------------------|--------------------------------------------------|
| September 1, 2013 - July 31, 2015 | Robotics Doctoral Curriculum Committee           |
| April 9, 2015                     | Michigan Robotics Day 2015 (co-organizer)        |
| April 10, 2014                    | Michigan Robotics Day 2014 (co-organizer)        |
|                                   | Host for speaker, Dr. Peter Wurman, Kiva Systems |

## Other Service Items at the University of Michigan

|                   |                                                       |
|-------------------|-------------------------------------------------------|
| November 15, 2013 | Engineering Graduate Symposium Poster Session (judge) |
|-------------------|-------------------------------------------------------|

## University of Toronto

### Committee Assignments and Memberships at the University of Toronto

|             |                                                                                    |
|-------------|------------------------------------------------------------------------------------|
| 2011        | UTIAS Director Search Advisory Board<br>Faculty of Applied Science and Engineering |
| 2009 - 2011 | Graduate Education and Research Committee<br>University of Toronto                 |
| 2009 - 2011 | UTIAS Student Experience Committee                                                 |
| April 2008  | Space Systems Design Contest (judge)<br>Faculty of Applied Science and Engineering |

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## Student Supervision, Mentoring, and Guidance

Co-supervisors are indicated in parentheses. The title of the thesis or project is given in *italics*.

## McGill University

### Postdoctoral Fellows and Research Associates Supervised at McGill University

1. Mr. Thomas Hitchcox, *Sensor Calibration and Uncertainty Characterization*, September 2023 - June 2024.  
Current position: Sonardyne Research Scientist - Navigation & Mapping.
2. Mr. Angad Bajwa, *Deep Inertial-Only Velocity Aided Estimation for Quadrotors*, September 2022 - August 2023.  
Current position: BAsC Student at the University of Waterloo.
3. Ms. Robyn Fortune, *Enhanced System Identification and Control of Large Scale Test Rigs*, May 2020 - October 2020.  
Current position: Engineer at CMLabs.
4. Mr. Alex Walsh, *Underwater SLAM Navigation*, May 2018 - April 2019.  
Current position: Guidance, navigation, and control engineering at Opener.
5. Ms. Eleonora Botta, *Enhancing the Capabilities of Space Situational Awareness*, January - April, 2018. September - December, 2018.  
Current position: Assistant Professor, Department of Mechanical and Aerospace Engineering, The University at Buffalo.

## Doctoral Students Supervised at McGill University

1. Ms. Desiree Fisker, *Multi-vehicle Teach and Repeat*, September 2025 - Present.
2. Mr. Jonathan Eid, *Robust Control of Power Systems*, September 2023 - Present.  
Direct transfer from MSc.
3. Ms. Yanting Zhou (J. Koveces), *Simulation of Contact*, September 2023 - Present.  
Recipient McGill Engineering Doctoral Award (MEDA). Direct transfer from MSc.
4. Mr. Arturo Del Castillo Bernal, *Underwater Navigation using Forward-looking Sonar, Cameras, and DVL-INS*, September 2023 - Present.  
Recipient McGill Engineering Doctoral Award (MEDA).
5. Mr. Vassili Korotkine, *Hessian Modifications in Nonlinear Least Squares*, September 2022 - Present.  
Recipient McGill Engineering Doctoral Award (MEDA).
6. Mr. Kyungmin Jung, *Outliner Rejection in Underwater SLAM*, September 2021 - Present.  
Recipient McGill Engineering Doctoral Award (MEDA).
7. Mr. Zi Cong Guo (T. Barfoot), *Navigation Using the Koopman Operator*, September 2021 - Present.  
Note: Mr. Guo is a PhD student at the University of Toronto that I unofficially co-advise with Prof. T. Barfoot.
8. Mr. Mitchell Cohen, *SLAM With Quadric Features*, September 2020 - Present.  
Recipient of a Vadasz Scholar McGill Engineering Doctoral Award (MEDA).
9. Mr. Steven Dahdah, *Linear Systems Theory and the Koopman Operator: Incorporating System Properties into Machine Learning for Dynamical Systems*, January 2020 - December 2024.  
Direct transfer from MEng (thesis).
10. Mr. Mohammed Shalaby (J. Le Ny), *Ultra-Wideband for Robot Navigation: Calibration, Communication, and State Estimation*, January 2021 - December 2023.  
Recipient of a McGill Engineering Doctoral Award (MEDA). Direct transfer from MEng (thesis).  
Current position: Robotics Engineer, Provectus Robotics Solutions.
11. Mr. Jonathan Arsenault, *Quantitative Analysis of Hockey Using Spatiotemporal Tracking Data*, January 2020 - December 2023.  
Recipient of a McGill Engineering Doctoral Award (MEDA).  
Current position: Data scientist, Carolina Hurricanes Hockey Club.
12. Mr. Thomas Hitchcox, *Strategies for Subsea Navigation and Mapping Using an Underwater Laser Scanner*, January 2019 - August 2023  
Recipient of a McGill Engineering Doctoral Award (MEDA).  
Current position: Sonardyne Research Scientist - Navigation & Mapping.
13. Mr. Charles Cossette (D. Saussié), *On-Manifold Estimation Methods for Multi-Robot Relative Localization using Ultra-Wideband Radio*, January 2019 - August 2023.  
Current position: Software engineer, Collision Avoidance Team, Zoox.
14. Ms. Qi Sun (J. Angeles), *Cartesian Elastodynamics Modelling of Parallel-kinematics Machines Under High-frequency, Small-amplitude Manoeuvres*, September 2016 - August 2023  
Recipient of a McGill Engineering Doctoral Award (MEDA).  
Current position: Associate Professor, College of Intelligent Systems Science and Engineering, Harbin Engineering University.
15. Mr. Daryoush Mirza Hekmati (W. Heath), *Control of Air-flow in Reciprocating Internal Combustion Engines*, University of Manchester, January 2015 - March 2019.  
Note: Mr. Hekmati was a PhD student at the University of Manchester that I unofficially co-supervise with Prof. W. Heath.

Current position: Senior Control Engineer at MDA Space, Spacecraft GNC system.

16. Mr. Stephen Alexander Chee, *Norm-Constrained State Estimation Strategies*, January 2013 - February 2018

Recipient of a McGill Engineering Doctoral Award (MEDA).

Current position: Mission Systems Engineering division of MDA Corporation.

17. Ms. Xiaowei Shan (J. Angeles), *Design, Prototyping and Testing of Biaxial MEMS Accelerometers for Rigid-body Pose-and-twist Estimation*, September 2012 - May 2017.

Recipient of a McGill Engineering Doctoral Award (MEDA).

Current position: Assistant Professor, School of Mechanical-Electronic and Vehicle Engineering, Beijing University of Civil Engineering and Architecture,.

18. Ms. Leila Bridgeman, *Methods Exploiting and Extending the Conic Sector Theorem*, January 2012 - October 2016.

Recipient of a McGill Engineering Doctoral Award (MEDA).

Current position: Assistant Professor, Dept. of Mechanical Engineering and Materials Science, Duke University.

### **Master of Science (Thesis) Students Supervised at McGill University**

1. Ms. Cayleigh Bates (J. Marshall), *Autonomous Mine Cleanup*, September 2026 - Present.
2. Mr. Charles Morain, *Multi-Robot Navigation*, September 2026 - Present.
3. Mr. Francis Maziade, *Passivity-Based Design of PHIL Interfaces*, September 2026 - Present.
4. Mr. Pierre Chamoun, *Multi-Sensor Calibration*, January 2026 - Present.
5. Mr. Mohammad Ghali (J. Bergthorson), *Control of Metal-Burning Fuel*, September 2025 - Present.
6. Ms. Isabel Loughheed, *Underwater Navigation*, September 2025 - Present.
7. Mr. Louis-Thomas Roy, *Off-Road Navigation*, September 2025 - Present.
8. Mr. Junha Yoo (D. Nwrouzezahrai), *Continuous-time State Estimation in Underwater Environments*, September 2024 - Present.
9. Mr. Andrew Stirling, *Machine Learning Approaches to UWB Noise Characterization*, September 2024 - Present.
10. Mr. Kyle Biron-Gricken, *Robust State Estimation using UWB*, September 2024 - Present.
11. Mr. Timothy Adams, *Robust Optimization Through the Lens of Control*, September 2024 - Present.
12. Mr. Nicholas Dahdah (J. Le Ny), *Multi-agent UWB Navigation*, January 2024 - Present.
13. Ms. Ashley Meagher, *Robust Control of Green Energy Systems*, September 2023 - Present.
14. Mr. Samuel Smocot (A. Sedal), *Observability Informed Optimal Sensor Placement for Soft Robots*, September 2023 - December 2025.
15. Mr. Sepehr Moalemi, *Input-Output Stability of First-Order Optimization Algorithms: A Passivity-Based Gain-Scheduling Approach*, September 2022 - December 2025.
16. Mr. Antoine Henri, *Construction and System Identification of a Hybrid Actuator*, September 2022 - August 2024.  
Current position: Control systems engineer, Kinova.
17. Mr. Louis Lortie, *Forward-backward Extended DMD with an Asymptotic Stability Constraint*, September 2022 - August 2024.  
Current position: PhD Student, McGill University.
18. Mr. Syed Shabbir Ahmed, *Range-aided Relative Pose Estimation and Formation Planning for Multi-robot Systems*, September 2022 - August 2024.

Winner of the 2025 Canadian Society for Mechanical Engineering Gold Medal for “Outstanding Academic Achievement in Mechanical Engineering.”

Current position: GNSS Algorithm Designer at Novatel Inc. - Hexagon Autonomy and Positioning

19. Mr. Philippe Decoste, *A Direct Photometric Approach to Homography Filtering using Thermal Imagery*, September 2021 - August 2024.
20. Mr. Arturo Del Castillo Bernal, *Bayesian Filtering for Homography Estimation*, September 2021 - August 2023.  
Current position: PhD Student, McGill University.  
Winner of the Michael P. Paidoussis Prize for the academic year 2022-2023.
21. Ms. Harriet Violet Chorney (M. Driscoll), *System Identification of Biological Tissue to Improve Haptic Interaction*, September 2022 - December 2023.  
Current position: Autonomy Developer, OTTO Motors by Rockwell Automation.
22. Ms. Natalia Pavlasek, *Simultaneous Localization and Mapping using Magnetic Fields and their Gradients*, January 2021 - August 2022.  
Current position: PhD Student, University of Washington.
23. Mr. Faraaz Ahmed, *Guidance and Model Predictive Control of a Tandem-rotor Helicopter*, September 2020 - August 2022.  
Current position: Quantitative Analysts, Financial Engineering & Modelling, Deloitte.
24. Mr. Vassili Korotkine, *State Estimation on Matrix Lie Groups using Iterative Filters*, September 2020 - August 2022.  
Current position: PhD Student, McGill University.
25. Mr. Daniil Lisus, *Dissecting Visual-Inertial SLAM: Errors, Landmarks, and Consistency*, September 2020 - August 2022.  
Current position: PhD Student, University of Toronto.
26. Mr. Kyungmin Jung, *Sliding-Window-Based Tightly-Coupled LIDAR-Inertial-Visual Odometry and Mapping*, September 2019 - December 2021.  
Current position: PhD Student, McGill University.
27. Mr. Amro Al-Baali, *Combining DVL-INS and Laser-based Loop Closures in a Batch Estimation Framework for Underwater Positioning*, September 2019 - December 2021.  
2020 - 2021 Best Seminar Award  
Current position: PhD Student, McGill University.
28. Mr. Marc-Antoin Lavoie, *Map-Constrained Train Navigation with IMU Measurements*, January 2019 - December 2021.  
Current position: PhD Candidate, University of Toronto.
29. Ms. Amanda Bianco (A.-M. Lauzon), *Automating Force Clamps to Measure Muscle Mechanics in Tissue Culture*, September 2018 - December 2020.  
Current position: Medical student, Montreal, QC.
30. Mr. Kevin Bergamin, *High-Performance Data-Driven Control of Physically-Based Human Characters*, September 2018 - August 2020. Current position: Graphics and Animation Specialist at Ubisoft
31. Mr. Mitchell Cohen, *Wind-Field Mapping Using a Coupled State and Wind-Velocity Estimator for Fixed-Wing Unmanned Aerial Vehicles*, September 2018 - August 2020.  
Current position: PhD student, McGill University.
32. Ms. Robyn Fortune, *Modern Control Approaches to Large-Scale Testing of Aerospace Structures*, January 2018 - December 2019.  
Current position: Engineer at CMLabs.



33. Mr. Ken Lee, *Negative Imaginary Controller Synthesis:  $\mathcal{H}_2$  and  $\mathcal{H}_\infty$  Approaches*, September 2017 - August 2019.  
Current position: Software engineer at Sanctuary AI.
34. Mr. Jonathan Arsenault, *Practical Considerations and Extensions of the Invariant Extended Kalman Filtering Framework*, September 2017 - August 2019.  
Current position: Data scientist, Carolina Hurricanes Hockey Club.
35. Mr. Zuyu Yin (J. Angeles), *Control of Fast Pick-and-Place Robots*, September 2016 - August 2018.  
Recipient of a McGill Engineering Doctoral Award (MEDA) and a Chinese Scholarship Council (CSC) award.  
Current position: Staff Engineer at CM Labs, Montreal, QC.
36. Mr. Duowen Qian, *Relative Feature Measurement Simultaneous Localization and Mapping in Three Dimensions*, September 2016 - December 2018. Current position: SLAM Expert at ARA Robotique, Montreal, QC.
37. Mr. Harsh Godbole, *Dynamic Modelling and Control of Cable-Actuated Systems*, April 2016 - October 2017.  
Current position: Aircraft Systems Software Specialist: Flight Controls, CAE, Montreal, QC.
38. Mr. Francois Hogan, *Dynamic Modelling and Control of Martian Tumbleweed Rovers*, September 2013 - February 2016.  
Current position: Facebook AI research.
39. Ms. Donya Mohammadshahi (M. Nahon), *The Dynamics and Control of Cable-Actuated Systems*, September 2011 - June 2013.  
Current position: MAYA HTT, Montreal, QC

### **Master of Engineering (Non-Thesis) Students Supervised at McGill University**

1. Mr. Matteo Hapiot, *Active Roll Control of an MRT Rocket*, April 2025 - December 2025.
2. Mr. Charles Cossette, *Adaptive Notch Filtering to Improve Attitude Estimation of UAVs*, April 2017 - April 2018.
3. Mr. Muniir Rohimun, *Passivity-Based Observer Design for Flexible Joint Manipulator Control*, September 2011 - August 2013.

### **Baccalaureate Students Supervised at McGill University**

#### **MECH 403/404 Honours Thesis**

1. Yorgo Abou Haidar (A. Higgins), *Attitude Control of a Balloon-Borne Payload*, January 2026 - December 2026.
2. Alexandre Hadji-Thomas, *Variational Inference for Robot Navigation*, January 2025 - December 2025.
3. Mr. Mikhail Kudryavtsev (M. Nahon), *Modelling and Stability Analysis of a Towed Echosounder*, January 2025 - December 2025.
4. Mr. Junha Yoo, *Coupled LiDAR-Visual-Inertial SLAM*, September 2023 - April 2024.
5. Mr. Kyle Biron-Gricken, *Anti-windup Control of UAVs*, September 2023 - April 2024.
6. Mr. Aldo Abou Chedid, *Magnetic Field Mapping Using Gaussian Process Regression*, January 2021 - December 2021.
7. Mr. Orlando Nicoletti, *Delayed-State Invariant Extended Kalman Filter for Monocular Visual-Inertial Navigation*, January 2019 - December 2019.

8. Mr. Amro Al-Baali, *Gain-Scheduling Conic-Sector-Based Tracking Control*, September 2017 - April 2018.
9. Mr. Sandro Papais, *Gravimeter-Enabled Interplanetary Navigation*, September 2017 - April 2018.
10. Mr. Dylan Caverly, *Gain-Scheduling Passivity-Based Tracking Control*, January 2016 - December 2016.
11. Mr. Alex Walsh, *Dynamic Modelling and Control of Flexible Robotic Space Manipulators with Telescoping Booms*, January 2012 - December 2012. **Best honours thesis winner.**
12. Mr. David Zlotnik, *A Comparison of Attitude Parameterizations for Spacecraft Attitude Control*, January 2012 - December 2012.
13. Mr. Francois Hogan, *Spacecraft Fuel Slosh: Modelling and Control*, January 2012 - December 2012.
14. Mr. E. Roger Fernando, *Dynamic Modelling and Control of Telescoping Manipulators with Lumped-Mass Flexibility*, September 2011 - April 2012.

### **MECH 498 Interdisciplinary Design Project 1**

1. Mr. Laurence Liang, *Machine Learning Classification of Channel Impulse Responses in UWB*, September 2024 - March 2025
2. Mr. Luis Sanchez, *Design of a Control Lab for MECH 412*, September 2022 - December 2022.
3. Mr. Samuel Herschmann, *Modelling and Control of Gyroelastic Systems*, January 2019 - April 2019.
4. Mr. Todd Scrimgeour, *Control Systems Design for Robotic Manipulator*, September 2017 - December 2017.

### **ECSE 478 Honours Thesis**

1. Irene Simier Requejo, *Radar Odometry*, September 2025 - April 2026.
2. Mr. Nicholas Dahdah, *Differentiable Filtering for Robot Navigation*, January 2023 - December 2023.
3. Mr. Titouan Delorme (Prof. Peter Caines), *Dynamic Power Cap and Regenerative Braking Control in a Formula Style Electric Racing Car*, September 2020 - April 2021.
4. Ms. Catherine Vanasse-Grosdidier, *Optimal Parallel Feedforward Control*, September 2018 - April 2019.
5. Mr. Sean Wolfe (Prof. Michael Rabbat), *UAV Target Tracking for AUVSI-SUAS Competition*, September 2016 - December 2016.

### **MECH 494 Honours Design Project**

1. Mr. Julien Otis-Laperrière, *Tandem-rotor Vehicle Design, Build, Fly*, September 2021 - December 2021.
2. Mr. Sandro Papais, *Control of Oxidizer Flow for the McGill Rocket Team Propulsion Systems*, September 2017 - April 2018.
3. Mr. Dylan Caverly, *Parallel Robot Manipulator Design*, January 2017 - April 2017.
4. Mr. Alex Walsh, *McHAB Descent Control*, January 2013 - April 2013.
5. Mr. David Zlotnik, *McHAB Attitude Control Hardware Design, Construction, and Testing*, January 2013 - April 2013.
6. Mr. Dylan Jude, *Desaturation Hardware for McHAB*, September 2012 - December 2012.

## Summer Students Supervised at McGill University

1. Mr. Didier Zhang, *AUV Thruster Design*, Design Team Internship, May - August 2026.
2. Ms. Natasha Lawford, *Subsea Vision Enhancement*, SURE, May - August 2026.
3. Ms. Cayleigh Bates (J. Marshall), *Autonomous Mine Cleanup*, SURE, May - August 2026.
4. Mr. Charles Morain, *Covariance Intersection Methods Applied to Multi-Robot Navigation*, SURE, May - August 2026.
5. Ms. Isabel Loughheed, *LBL SLAM when Heading is Known, Part 2*, SURE, May - August 2025.
6. Ms. Irene Simier Requejo, *Radar-IMU Odometry*, SURE, May - August 2025.
7. Mr. Mohammad Ghali (J. Bergthorson), *Control of Metal-Burning Fuel*, SURE, May - August 2025.
8. Mr. Angad Bajwa, *Benchmarking Feature Detection Underwater*, Summer Project, May - August 2025.
9. Ms. Isabel Loughheed, *LBL SLAM when Heading is Known, Part 1*, SURE, May - August 2024.
10. Mr. Junha Yoo, *Continuous-time State Estimation*, SURE, May - August 2024.
11. Mr. Andrew Stirling, *Machine Learning Approaches to UWB Noise Characterization*, SURE, May - August 2024.
12. Mr. Kyle Biron-Gricken, *Robust State Estimation using UWB*, SURE, May - August 2024.
13. Mr. Timothy Adams, *Robust-Performance Observer Design*, SURE, May - August 2024.
14. Mr. Jonathan Eid, *Implementing Antiwindup Compensation in Linear Control Systems*, SURE, May - August 2023.
15. Mr. Samuel Smocot (A. Sedal), *Controllability and Observability of Soft Robots*, SURE, May - August 2023.
16. Mr. Nicholas Dahdah, *Learning Noise Characteristics of Object Detections for Multi-Object Tracking*, SURE, May - August 2023.
17. Mr. Syed Shabbir Ahmed, *Formation and Attitude Control of Decentralized Multi-robot Systems*, SURE, May - August 2022.
18. Mr. Louis Lortie, *Koopman-based System ID*, May - August 2022.
19. Ms. Jiaying Fang, *Experimental Evaluation of Teach and Repeat with UWB*, May - August 2022.
20. Mr. Jasper Yun, *Benchmark Dataset Collection*, SURE, NSERC USRA recipient, May - August 2021.
21. Ms. Kattly Li, *Benchmark Dataset Collection*, SURE, May - August 2021.
22. Ms. Natalia Pavlasek, *An Invariant Extended Kalman Filter for Navigation using Two Position Receivers and an Inertial Measurement Unit*, SURE, NSERC USRA recipient, May - August 2020.
23. Mr. Daniil Lisus, *Learning Position and Orientation of a Robot Using Ultra-wideband Antenna Received Signal Strength*, SURE, NSERC USRA recipient, winner of the best SURE project, General and Multidisciplinary category, May - August 2020.
24. Mr. Niels van der Laan, *Invariant RTS Smoothing*, GERAD Sponsored Student, May - August 2019.
25. Mr. Orlando Nicoletti, *Monocular Visual-Inertial Navigation*, SURE, May - August 2019.
26. Mr. Kyungmin Jung, *Autonomous Navigation of Robotic Systems Using the Invariant Sliding Window Filter*, SURE, May - August 2019.
27. Mr. Amro Al-Baali, *Comparing Exteroceptive Sensors for Autonomous Underwater Vehicle Navigation*, SURE, May - August 2019.
28. Mr. Raphael Zonis (Dr. G. Ijpma and Prof. A.-M. Lauzon), *Modelling, System Identification, and Control of Active Viscoelastic Tissues*, SURE, May - August 2018.
29. Mr. Niels van der Laan, *Fatigue Structural Testing Enhancement Research (FASTER)*, SURE, May -

August 2018.

30. Mr. Mitchell Cohen, *Modelling and Control of a VTOL to Forward-flight UAV*, SURE, May - August 2018.
31. Mr. Martin Doff-Sotta, *Hardware in the Loop Testing for a VTOL to Forward-flight UAV*, May - August 2018.
32. Mr. Shatil Rahman, *Simultaneous Localization and Mapping as a Probabilistic Constrained Optimization Problem*, SURE, May - August 2018.
33. Mr. Rashik Habib, *Variance-Constrained Control using Gaussian Process Models*, SURE, May - August 2018.
34. Mr. Paul Albert-Lebrun, *High-altitude Balloon Communications*, MIAE-funded student project, May - August 2017.
35. Mr. Robin Aucoin, *LMI-Constrained Kalman Filtering*, SURE, NSERC USRA recipient, May - August 2017.
36. Mr. Jonathan Arsenault, *Autonomous Parafoil Navigation, Guidance, and Control*, SURE, NSERC USRA recipient, winner of the best SURE project, Aerospace Engineering category, May - August 2017.
37. Ms. Lindsey Grant, *Parafoil Modelling and Control*, May - August 2017.
38. Mr. Cornelius Hart, *Implementation of a Kalman Filter on a Google Tango Tablet*, SURE, MIAE co-funded, May - August 2016.
39. Mr. Anthony Maalouly, *Nyquist Interpretation of the Small Gain, Large Gain, and Passivity Theorems*, SURE, MIAE co-funded, May - August 2016.
40. Mr. Dylan Caverly, *Control of a Flexible-Link Manipulator with Zero Steady-State Tracking Error*, SURE, NSERC USRA recipient, MIAE co-funded, winner of the best SURE project, Aerospace Engineering category, May - August 2016.
41. Mr. David Zlotnik (C. A. Beltempo), *Control Systems Enhancement for Large Scale Testing* (Project with the National Research Council (NRC) of Canada), May - August 2013.
42. Mr. Paul Aubin, *McHAB Descent Control Design, Construction, and Test* (visiting student from Supaero, France), May - August 2013.
43. Mr. Nguyen Khoi Tran, *Attitude Control for a Stratospheric Balloon Platform*, SURE, MIAE co-funded, May - August 2013.
44. Mr. Ryan Caverly, *Dynamic Modelling and Passivity-Based Control of Cable-Actuated Systems* SURE, NSERC USRA recipient, May - August 2013.
45. Mr. Dylan Caverly, *Computer Aided Design of McHAB Payload and Cable-Robot Testbed*, May - August 2013.
46. Mr. Mohamed Najih, *Using an Observer to Synthesize Optimal Strictly Positive Real and Conic Controllers*, SURE, NSERC USRA recipient, May - August 2013.
47. Mr. Francois Hogan, *Dynamic Modelling and Stability Analysis of a Martian Tumbleweed Rover*, SURE, NSERC USRA recipient, MIAE co-funded, winner of the best SURE project, Aerospace Engineering category, May - August 2013.
48. Mr. David Zlotnik, *Flexible Spacecraft Modelling and Control*, SURE, NSERC USRA recipient, MIAE co-funded, May - August 2012.

## Student Research Projects Supervised at McGill University

1. Mr. Tim Rietberg, *System Identification of a Metal Burning Process*, Visiting MSc Student from TU Eindhoven, September - December 2025.
2. Mr. Philip Becker, *Control of a Heat Pump and Water Heating System*, Independent Project, April - August 2025.
3. Mr. Dorian Bernard, *Control of an Autonomous Catamaran*, Capstone Design Project, January 2025-April 2025. (Mr. Bernard is an ECE student from Polytechnique Montréal.)
4. Mr. Alex Beaudin, *LiDAR-Camera-IMU Head Construction*, Independent Project, Sept. 2020 - April 2021.
5. Mr. Julien Otis-Laperriere, *Research and Experimental Testing of Liquid-Injection Thrust Vector Control Actuator (LITVC)*, Mitacs Accelerate, August 2019 - December 2019.
6. Mr. Sandro Papais, *Launch Vehicle Design and Trajectory Optimization Using Direct Collocation and Nonlinear Programming*, NSERC Engage, February - May, 2019.

## Student Teams and Projects Supervised at McGill University

- |                                                                                                                                                                                                                                                            |             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1. <i>Design of a Semi-Active Suspension System for an FSAE Electric Race Car</i><br>Mechanical Engineering Capstone Project, MECH 463<br>Eric Masselot, Estelle Huy, Rocco Racioppo, Jonas Maarek                                                         | 2025 - 2026 |
| 2. <i>Design of a Folding Arm Mechanism for a Drone</i><br>Mechanical Engineering Capstone Project, MECH 463<br>Charles Morin, Karim Saliba, Fouad Al Laham, Salman Dajani                                                                                 | 2025 - 2026 |
| 3. <i>Design of Liquid Cooling Controller for an FSAE Electric Race Car</i><br>Mechanical Engineering Capstone Project, MECH 463 (co-advised with D. Caverly)<br>Philip Becker, Francois-Xavier Duclos, Shridhar Baichoo, Marina Pereira Garcia            | 2024 - 2025 |
| 4. <i>Design of a Towed Echosounder</i><br>Mechanical Engineering Capstone Project, MECH 463 (co-advised with M. Nahon)<br>Mr. Karim Al Sabbagh, Mr. Frédéric Vallée, Mr. Nicolas Thirion, Mr. Auriano Julsain.                                            | 2024 - 2025 |
| 5. <i>Control System for McGill Formula Electric</i> (3rd place at Design Day)<br>Mechanical Engineering Capstone Project, MECH 463<br>Ms. Kattly Li, Mr. Philippe Voyer, Mr. Jeremy Rose, Mr. Louis Lortie                                                | 2021 - 2022 |
| 6. <i>Autonomous Train Testbed</i><br>Mechanical Engineering Capstone Project, MECH 463<br>Ms. Carly Bennett, Mr. Christopher Chung, Ms. Emily Stotland, Mr. Daniel Schwartz                                                                               | 2020 - 2021 |
| 7. <i>Spinning LIDAR for UAV Navigation with ARA Robotique</i> (3rd place at Design Day)<br>Mechanical Engineering Capstone Project, MECH 463<br>Mr. Maksim Solovyev, Mr. Victor Rolland, Mr. Andrej Jermilov, Mr. Qitong Zhang                            | 2020 - 2021 |
| 8. <i>Development of a Control System for a FSAE Electric Race Car</i><br>Mechanical Engineering Capstone Project, MECH 463<br>Mr. Ali Al-taher, Mr. Sresht Annadevara, Mr. Félix Couturier-Lamy, Mr. John Jung                                            | 2018 - 2019 |
| 9. <i>Design of a Planar Parallel Robot for Inertia-Parameter Identification of a Rigid-Body</i><br>Mechanical Engineering Capstone Project, MECH 463<br>Co-supervised with Prof. J. Angeles<br>Mr. Omar Mahmoud, Mr. Phillip Salamy, Mr. Mohammed Shalaby | 2018 - 2019 |
| 10. <i>Unmanned Aerial Vehicle Deployment from a Tethered Balloon - Team 1</i>                                                                                                                                                                             | 2017 - 2018 |

- Mechanical Engineering Capstone Project, MECH 463  
Ms. Jessica Ronci, Mr. Nicholas Giroux-Lacombe, Mr. Jean-Christophe Maranda, Mr. Benoit Labelle
11. *Unmanned Aerial Vehicle Deployment from a Tethered Balloon - Team 2* 2017 - 2018  
Mechanical Engineering Capstone Project, MECH 463  
Mr. Mitchell Cohen, Mr. Robert Joy, Ms. Catherine Koutsaris, Mr. Alessandro Defazio
  12. *Using Order to Reduce People-Elephant Conflict* 2017 - 2018  
Mechanical Engineering Capstone Project, MECH 463  
Mr. Ted Li, Ms. Yiru Wang, Mr. Ryan Dotsikas, Mr. Nick Brunt.
  13. *Hybrid Rocket Engine Development* 2017 - 2018  
Mechanical Engineering Capstone Project, MECH 463  
Mr. Jonathan Lesage, Mr. Austin L'Ecuyer, Mr. Kyle Weissman, Mr. Thomas Hannaford
  14. *Desktop Photogrammetry - A 3D Scanning Approach* 2017 - 2018  
Electrical Engineering Capstone Project, ECSE 474/475  
Mr. Thomas Faribault-Ménard, Mr. Gabriel Chootong, Mr. Nicholas Roy, Mr. Jeslan Rajendram
  15. *McGill Cubesat ACDS Design* 2017 - 2018  
Electrical Engineering Capstone Project, ECSE 456  
Mr. Roberto Casoli, Mr. Eric Haniak, Mr. Philippe Lessard
  16. *McGill Cubesat ACDS Design* 2016 - 2017  
Mechanical Engineering Capstone Project, MECH 463  
Mr. Talha Tariq, Mr. Xiaokang Deng, Mr. Calvin Liao, Mr. Charles Turner
  17. *The McGill High Altitude Balloon (McHAB) Project* 2012 - 2013  
Electrical Engineering Capstone Project, ECSE 474/475  
Mr. Nguyen Khoi Tran, Mr. Nam Nguyen, Mr. Long Ma, and Mr. Xiao He
  18. *2012 CANSAT McGill Team* 2011 - 2012  
Mechanical Engineering Capstone Project, MECH 463  
Ms. Laura Richardson, Mr. Marc Gaulier, Mr. Lucas Kruitwagen, Mr. Rory Ewing, Mr. Umaid Imran, and Mr. Nguyen Khoi Tran

## University of Michigan

### Doctoral Students Supervised at the University of Michigan

1. Mr. David Evan Zlotnik, *Lie Group Observer Design for Robotic Systems: Extensions, Synthesis, and Higher-Order Filtering*, University of Michigan, January 2014 - August 2018.  
Current position: Autonomous vehicles engineer at Zoox.
2. Mr. Ryan James Caverly, *Optimal Output Modification and Robust Control using Minimum Gain and the Large Gain Theorem*, University of Michigan, January 2014 - March 2018.  
Current position: Assistant Professor, Dept. of Aerospace Engineering and Mechanics, University of Minnesota.
3. Mr. Alex Walsh, *Extremum-Seeking Guidance and Conic-Sector-Based Control of Aerospace Systems*, University of Michigan, September 2013 - December 2017.  
Current position: Guidance, navigation, and control engineering at Opener.

### Master of Engineering (Non-Thesis) Students Supervised at the University of Michigan

1. Mr. Phillip Hargrove, *Flexible Spacecraft Modelling*, University of Michigan, September 2014 - December 2014.

2. Mr. Anand Swaminathan, *Modelling of a Spherical Rover*, University of Michigan, September 2013 - April 2014.  
Current position: Control Systems Engineer, Tesla Motors, Palo Alto, CA.
3. Mr. Evan Hilgemann, *Pointing Control of Small Balloon Payloads*, University of Michigan, September 2013 - April 2014.  
Current position: Engineer, Jet Propulsion Lab, Pasadena, CA.

## Baccalaureate Students Supervised at the University of Michigan

### AERO 390 Directed Study

1. Mr. Ryan Patterson, *Dynamic Modelling and Control of Thin Floating Plates*, January 2015 - April, 2015
2. Mr. Jin Woo Park, *Angular Acceleration and Velocity Estimation Using Accelerometer Strapdowns*, January 2015 - April, 2015
3. Mr. Roshan Radhakrishnan, *Balloon Gondola Control Systems Design and Hardware Implementation*, September 2014 - April 2015.

### Summer Students

1. Mr. Ryan Patterson, *Robust Control of Thin Floating Plates*, May - August, 2015
2. Mr. Jin Woo Park, *Kalman Filtering of Measurements used in Accelerometer Strapdowns*, May - August, 2015
3. Mr. Dylan Caverly, *Attitude Estimation Hardware Construction*, July - August, 2014.
4. Mr. Andrew Taylor (Prof. J. Cutler), *Strato Estimator Design*, May - August, 2014.
5. Mr. Roshan Radhakrishnan (Prof. J. Cutler), *Strato Control Systems Design*, May - June, 2014.
6. Mr. David Carter (Prof. J. Cutler), *Strato Dynamics*, May - June, 2014.

## Student Teams and Projects Supervised at the University of Michigan

1. *MiTEE - Miniature Tether Electrodynamics Experiment* (Prof. B. Gilchrist) 2014 - 2015  
10 to 12 team members annually consisting of undergraduate, M.Eng., and Ph.D. students from UMich tasked with designing and prototype testing a femto- to nano-satellite size electrodynamic tether.
2. *Strato Balloon Team* (Prof. J. Cutler) 2013 - 2015  
10 to 12 team members annually consisting of undergraduate, M.Eng., and Ph.D. students from UMich tasked with designing, building, testing, and launching a high-altitude balloon system.

## Pro-dean at McGill University

1. Mr. Joey Litalien November 18, 2024  
Dept. of Electrical and Computer Engineering
2. Ms. Kamila Mustafina June 16, 2023  
Dept. of Chemistry
3. Ms. Gillian Klassen September 23, 2022  
Dept. of Educational & Counselling Psychology
4. Mr. Shuang Gao November 2018  
Dept. of Electrical and Computer Engineering

## Doctoral Candidate Committees at McGill University

As a member of a doctoral committee at McGill University, thesis reading and evaluation is compulsory, as is attending the student's defense.

- |                                                                 |                                                      |
|-----------------------------------------------------------------|------------------------------------------------------|
| 1. Mr. Louis Lortie                                             | April 2025 - Present                                 |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Guillaume Durandau                 |
| 2. Mr. Yuanyuan Tao                                             | Feb. 2025 - Present                                  |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Profs. Paul Kry and Derek Nowrouzezahrai |
| 3. Mr. Daniel Kong                                              | Sept. 2023 - Present                                 |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Profs. Paul Kry and Derek Nowrouzezahrai |
| 4. Mr. Wilfred Mason                                            | Sept. 2022 - Present                                 |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Audrey Sedal                       |
| 5. Mr. Celia Hameury                                            | Sept. 2022 - Present                                 |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Marco Amabili                      |
| 6. Mr. Yaojie Cai                                               | Jan. 2021 - Present                                  |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Prof. Xiaozhe Wang                       |
| 7. Mr. Xingshuai Huang                                          | Jan. 2021 - Present                                  |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Prof. Benoit Boulet                      |
| 8. Mr. Xu Dai                                                   | Sept. 2020 - Present                                 |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Jozsef Kovecses                    |
| 9. Mr. Ali Raoofian                                             | Sept. 2019 - Present                                 |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Jozsef Kovecses                    |
| 10. Ms. Fares El Tin                                            | Sept. 2019 - May 2022                                |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Meyer Nahon                        |
| 11. Ms. Juan Carlos Hernández Ramirez                           | Sept. 2019 - Present                                 |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Meyer Nahon                        |
| 12. Ms. Sally Zabetian-Hosseini                                 | January 2020 - Present                               |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisors: Profs. Benoit Boulet and Geza Joos      |
| 13. Mr. Ehsan Dehghanpourl                                      | January 2019 - Present                               |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Prof. Geza Joos                          |
| 14. Ms. Shrouti Dutta                                           | September 2019 - Present                             |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Arun Misra                         |
| 15. Mr. Hamed Layeghi                                           | January 2017 - Present                               |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Prof. Peter Caines                       |
| 16. Mr. Syam Vangar                                             | January 2017 - Present                               |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Siva Nadarajah                     |
| 17. Mr. Qingfeng Lou                                            | March 2017 - Present                                 |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Jozsef Kovecses                    |
| 18. Ms. Isabelle Jean                                           | April 2016 - April 2019                              |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Arun Misra                         |
| 19. Mr. Majid Sheikholeslami                                    | September 2011 - Present                             |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Jozsef Kovecses                    |



|                                                                 |                                   |
|-----------------------------------------------------------------|-----------------------------------|
| 20. Mr. Avishek Paul                                            | January 2016 - August 2018        |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Prof. Geza Joos       |
| 21. Mr. Albert Peiret Gimenez                                   | January 2016 - August 2019        |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Jozsef Kovecses |
| 22. Mr. Amir Hossein Rabbani                                    | September 2011 - September 2016   |
| School of Computer Science, McGill University                   | Supervisor: Prof. Paul Kry        |
| 23. Mr. Ali Pakniyat                                            | September 2011 - September 2016   |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Prof. Peter Caines    |
| 24. Mr. Farnood Gholami                                         | September 2011 - May 2016         |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Jozsef Kovecses |
| 25. Mr. Ying Yu                                                 | September 2012 - April 2017       |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Luc Mongeau     |
| 26. Mr. Waqas Khan                                              | September 2012 - August 2016      |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Meyer Nahon     |
| 27. Mr. Sheldon Andrews                                         | September 2012 - August 2013      |
| School of Computer Science, McGill University                   | Supervisor: Prof. Paul Kry        |

### Doctoral Theses Reviewed As Internal Examiner at McGill University

As an internal examiner at McGill University, reading the student's thesis and writing of a comprehensive evaluation report is required.

|                                                                 |                                     |
|-----------------------------------------------------------------|-------------------------------------|
| 1. Mr. Yaojie Cai                                               | June 2025                           |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Prof. Xiaozhe Wang      |
| 2. Mr. Xu Dai                                                   | April 2024                          |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Jozsef Kovecses   |
| 3. Mr. Chen Chen                                                | September 2023                      |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Jozsef Kovecses   |
| 4. Ms. Imen Jendoubi                                            | September 2022                      |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Prof. Francois Bouffard |
| 5. Mr. Ahmad Haidar                                             | March 2013                          |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Prof. Benoit Boulet     |
| 6. Ms. Nona Abolfathi Nobari                                    | February 2013                       |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Arun Misra        |
| 7. Mr. Farzin Taringoo                                          | September 2012                      |
| Dept. of Electrical and Computer Engineering, McGill University | Supervisor: Prof. Peter Caines      |
| 8. Mr. Torsten Liesk                                            | April 2012                          |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Meyer Nahon       |
| 9. Mr. Danial Alizadeh                                          | March 2012                          |
| Dept. of Mechanical Engineering, McGill University              | Supervisor: Prof. Jorge Angeles     |

### Master Theses Reviewed As Internal Examiner at McGill University

As an internal examiner at McGill University, reading the student's thesis and writing of a comprehensive evaluation report is required.

- |                                                                                                              |                                                                           |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1. Ms. Reihaneh Ghoroghchian, M.Sc.<br>Dept. of Electrical Engineering, McGill University                    | September 2023<br>Supervisor: Prof. Peter Caines and Prof. Aditya Mahajan |
| 2. Mr. Daniel Kong, M.Sc.<br>School of Computer Science, McGill University                                   | September 2023<br>Supervisor: Prof. Paul Kry                              |
| 3. Mr. Jad Wehbeh, M.Sc.<br>Dept. of Mechanical Engineering, McGill University                               | June 2022<br>Supervisor: Prof. Inna Sharf                                 |
| 4. Mr. Jack Minnetian, M.Sc.<br>Dept. of Electrical Engineering, McGill University                           | September 2021<br>Supervisor: Prof. Derek Nowrouzezahrai                  |
| 5. Mr. Sahand Rezaei-Shoshtari, M.Eng. (Thesis)<br>Dept. of Mechanical Engineering, McGill University        | February 2020<br>Supervisor: Prof. Inna Sharf                             |
| 6. Mr. Patrick Abouzakhm, M.Eng. (Thesis)<br>Dept. of Mechanical Engineering, McGill University              | April 2016<br>Supervisor: Prof. Inna Sharf                                |
| 7. Ms. Mirunalini Thirugnanasambandam, M.Eng. (Thesis)<br>Dept. of Mechanical Engineering, McGill University | April 2012<br>Supervisor: Prof. Mathias Legrand                           |
| 8. Ms. Wing Yan Edith Mak, M.Eng. (Non-Thesis)<br>Dept. of Mechanical Engineering, McGill University         | January 2012<br>Supervisor: Prof. Jorge Angeles                           |
| 9. Mr. Francis Bellefeuille, M.Eng. (Thesis)<br>Dept. of Mechanical Engineering, McGill University           | December 2011<br>Supervisor: Prof. Arun Misra                             |
| 10. Mr. Florian Shkurti, M.Sc.<br>School of Computer Science, McGill University                              | October 2011<br>Supervisor: Prof. Gregory Dudek                           |

### Doctoral Candidate Committees at the University of Michigan

As a member of a doctoral committee at the University of Michigan, thesis reading, thesis evaluation including a report, and defence attendance is compulsory.

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|------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 1. Mr. Frantisek Sobolic<br>Dept. of Aerospace Engineering, University of Michigan | September 2013 - August 2015<br>Supervisor: Prof. Dennis Bernstein   |
| 2. Mr. Khaled Aljanaideh<br>Dept. of Aerospace Engineering, University of Michigan | September 2013 - May 2015<br>Supervisor: Prof. Dennis Bernstein      |
| 3. Mr. Gerardo Cruz<br>Dept. of Aerospace Engineering, University of Michigan      | September 2013 - December 2014<br>Supervisor: Prof. Dennis Bernstein |
| 4. Mr. Justin Rufa<br>Dept. of Aerospace Engineering, University of Michigan       | September 2013 - July 2014<br>Supervisor: Prof. Ella Atkins          |

### Doctoral Theses Reviewed As External Examiner for Other Institutions

Thesis reading, thesis evaluation including a report, is compulsory. In some cases, attending the defence in person or online was required.

- |                                                                                    |                                                 |
|------------------------------------------------------------------------------------|-------------------------------------------------|
| 1. Mr. Kevin Course<br>University of Toronto                                       | January 2025<br>Supervisor: Prof. Prasanth Nair |
| 2. Mr. Tarek Bouazza<br>Université Côte D'Azur                                     | December 2024<br>Supervisor: Prof. Tarek Hamel  |
| 3. Mr. Roland Jung<br>Control of Networked Systems Group, University of Klagenfurt | October 2023<br>Supervisor: Prof. Stephan Weiss |

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|---------------------------------------------------------------------|--------------------------------------------------------|
| 4. Mr. Seyed Mohammadmoein Mousavi                                  | August 2023                                            |
| Dept. of Chemical Engineering, Queen's University                   | Supervisor: Prof. Martin Guay                          |
| 5. Mr. Reza Babazadeh                                               | December 2021                                          |
| Dept. of Electrical and Computer Engineering, Concordia University  | Supervisor: Prof. Rastko R. Selmic                     |
| 6. Mr. Meysam Effati                                                | June 2020                                              |
| Dept. of Electrical and Computer Engineering, Concordia University  | Supervisor: Prof. Krzysztof Skonieczny                 |
| 7. Mr. Paul Chauchat                                                | January 2020                                           |
| Mines ParisTech, PSL Research University                            | Supervisor: Prof. Silvère Bonnabel and Dr. Axel Barrau |
| 8. Mr. Hossein Beikzadeh                                            | January 2014                                           |
| Dept. of Electrical and Computer Engineering, University of Alberta | Supervisor: Prof. Horacio Marquez                      |

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## **Teaching and Instruction**

**McGill University**

**University of Michigan**

**University of Toronto (as a Teaching Assistant)**

| Course Number | Title                                       | Level                  | Semester    | Enrollment |
|---------------|---------------------------------------------|------------------------|-------------|------------|
| MECH 220      | Mechanics 2                                 | Undergraduate          | Winter 2012 | 118        |
| MECH 309      | Numerical Methods in Mech. Eng.             | Undergraduate          | Winter 2019 | 91         |
|               |                                             |                        | Fall 2021   | 88         |
|               |                                             |                        | Fall 2022   | 89         |
|               |                                             |                        | Fall 2023   | 91         |
|               |                                             |                        | Fall 2024   | 91         |
| MECH 412      | System Dynamics and Control                 | Undergraduate          | Fall 2015   | 63         |
|               |                                             |                        | Winter 2016 | 99         |
|               |                                             |                        | Winter 2017 | 83         |
|               |                                             |                        | Fall 2017   | 75         |
|               |                                             |                        | Winter 2018 | 95         |
|               |                                             |                        | Fall 2018   | 76         |
|               |                                             |                        | Winter 2021 | 105        |
|               |                                             |                        | Winter 2022 | 114        |
|               |                                             |                        | Fall 2023   | 67         |
|               |                                             |                        | Fall 2025   | 55         |
| MECH 403/404  | Honours Thesis                              | Undergraduate          | Fall 2025   | 22         |
|               |                                             |                        | Winter 2026 | XX         |
| MECH 513*     | Control Systems (Linear Systems)            | Undergraduate/Graduate | Fall 2012   | 33         |
|               |                                             |                        | Fall 2016   | 44         |
|               |                                             |                        | Winter 2021 | 23         |
|               |                                             |                        | Winter 2022 | 26         |
|               |                                             |                        | Winter 2023 | 31         |
|               |                                             |                        | Winter 2024 | 35         |
|               |                                             |                        | Winter 2025 | 16         |
|               |                                             |                        | Fall 2025   | 49         |
| MECH 542      | Spacecraft Dynamics                         | Undergraduate/Graduate | Fall 2012   | 21         |
| MECH 642      | Advanced Dynamics                           | Graduate               | Winter 2019 | 23         |
|               |                                             |                        | Winter 2025 | 18         |
| MECH 672      | Estimation and Control of Rob. & Aero. Sys. | Graduate               | Winter 2016 | 10         |
|               |                                             |                        | Winter 2018 | 18         |
|               |                                             |                        | Fall 2020   | 9          |

Table 1: Undergraduate and graduate courses taught at McGill University. Courses listed in parentheses are cross listed. \*Cross-listed with ECSE 501 in 2012 & 2016.

| Course Number       | Title                                         | Level         | Semester    | Enrollment |
|---------------------|-----------------------------------------------|---------------|-------------|------------|
| AERO 347            | Space Flight Mechanics                        | Undergraduate | Winter 2014 | 45         |
|                     |                                               |               | Winter 2015 | 44         |
| AERO 540 (MECH 540) | Intermediate Dynamics (Intermediate Dynamics) | Graduate      | Fall 2014   | 67         |

Table 2: Undergraduate and graduate courses taught at the University of Michigan. Courses listed in parentheses are cross listed.

| Course Number | Title                                  | Level                  | Semester                   |
|---------------|----------------------------------------|------------------------|----------------------------|
| ECE 410       | Control Systems                        | Undergraduate          | Fall 2011                  |
| AER 301       | Classical Mechanics                    | Undergraduate          | Fall 2008<br>Fall 2009     |
| AER 525       | Robotics                               | Undergraduate/Graduate | Fall 2006<br>Fall 2007     |
| AER 1513      | Spacecraft Dynamics and Control 2      | Graduate               | Winter 2011                |
| APS 1003      | Professional Education and Instruction | Graduate               | Summer 2009<br>Summer 2010 |

Table 3: Undergraduate and graduate courses TAed at the University of Toronto.

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## Industrial Work Experience and Consulting

|                                                |                                                                                                                                                                                                                                               |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| November 2024 - Present                        | <i>Sonardyne International</i> , Yateley, UK<br>Job title: Strategic Advisor.<br>Job description: Provide guidance on various aspects of subsea navigation.                                                                                   |
| 2019                                           | <i>Samsung Artificial Intelligence Center</i> , Montreal, QC<br>Job title: Consultant.<br>Job description: Consulted on various aspects of robot navigation.                                                                                  |
| November 2016                                  | <i>Kitty Hawk Aerospace</i> , Palo Alto, CA<br>Job title: Consultant.<br>Job description: Consulted on various aspects of UAV navigation and control.                                                                                         |
| May - August 2013,<br>February 2010, June 2009 | <i>G &amp; M Engineering</i> , Burlington, ON<br>Job title: Consultant.<br>Job description: Structural and dynamic analysis of automated material handling equipment.                                                                         |
| April - August 2006                            | <i>National Steel Car</i> , Hamilton, ON<br>Job title: Design engineer and finite element modelling analyst.<br>Job description: Finite element modelling/analysis and optimization of railway car structures, substructures, and components. |
| September - December 2005                      | <i>MechDes Engineering</i> , Nunspeet, The Netherlands<br>Job title: Machine design engineering student.<br>Job description: Design, simulation, and modelling of automated welding equipment.                                                |
| January - April 2005                           | <i>Nutech Engineering</i> , Milton, ON<br>Job title: Automation engineering student.<br>Job description: PLC programming and electrical system design including motion control, fault and safety logic.                                       |
| November 2004                                  | <i>Abtron Automation</i> , Stratford, ON<br>Job title: Automation consultant.<br>Job description: Design of software algorithm to determine tool-path of five axis CNC machine.                                                               |
| May - August 2004                              | <i>Burlington Automation</i> , Burlington, ON<br>Job title: Machine design engineering student.<br>Job description: Designed and built an automated material handling, measuring, and cutting system.                                         |
| September - December 2003                      | <i>Thru-Way Trailer</i> , Milton, ON<br>Job title: Manufacturing engineering student.<br>Job description: Commissioning of CNC plasma cutting machines. Design of automated welder to manufacture flat-bed trailers.                          |
| January - April 2003                           | <i>National Steel Car</i> , Hamilton, ON<br>Job title: Jig design engineering student.<br>Job description: Design of railway car jigs and fixtures for railway car manufacturing.                                                             |

April - August 2002

*Eldorado Tool and Manufacturing, Waterloo, ON*

Job title: General machinist and welding student.

Job description: Manufacturing of parts using manual and CNC lathes and milling machines. Arc (SMAW) and Mig (GMAW) welding, oxy-acetylene cutting, fixture design.

1999 - 2004

*Spokes and Slopes, Milton, ON*

Job title: Bicycle mechanic, ski technician, sales representative.

Job description: Certified bike mechanic (Level 1) responsible for all aspects of bicycle repair and maintenance.

### **Industry Related Software Skills**

Computer Aided Design: Solid Works, Solid Edge, AutoCAD, Inventor, ProEngineer.

Computer Programming: python, matlab, Visual Basic, C, C++, Pascal, Delphi, MathCAD.

General: L<sup>A</sup>T<sub>E</sub>X, Microsoft products.